



Module 11: IPv4 Addressing

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Introduction to Networks v7.0
(ITN)





Module 11: IPv4 Addressing

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Module Objectives

Module Title: IPv4 Addressing

Module Objective: Calculate an IPv4 subnetting scheme to efficiently segment your network.

Topic Title	Topic Objective
IPv4 Address Structure	Describe the structure of an IPv4 address including the network portion, the host portion, and the subnet mask.
IPv4 Unicast, Broadcast, and Multicast	Compare the characteristics and uses of the unicast, broadcast and multicast IPv4 addresses.
Types of IPv4 Addresses	Explain public, private, and reserved IPv4 addresses.
Network Segmentation	Explain how subnetting segments a network to enable better communication.
Subnet an IPv4 Network	Calculate IPv4 subnets for a /24 prefix.

Module Objectives (Cont.)

Module Title: IPv4 Addressing

Module Objective: Calculate an IPv4 subnetting scheme to efficiently segment your network.

Topic Title	Topic Objective
Subnetting a /16 and a /8 Prefix	Calculate IPv4 subnets for a /16 and a /8 prefix.
Subnetting to Meet Requirements	Given a set of requirements for subnetting, implement an IPv4 addressing scheme.
Variable Length Subnet Masking (VLSM)	Explain how to create a flexible addressing scheme using variable length subnet masking (VLSM).
Structured Design	Implement a VLSM addressing scheme.

11.1 IPv4 Address Structure

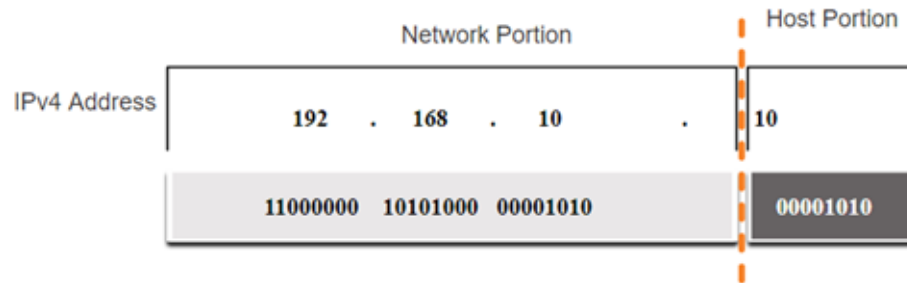
IPv4 Address Structure

Network and Host Portions

An IPv4 address is a 32-bit hierarchical address that is made up of a network portion and a host portion.

When determining the network portion versus the host portion, you must look at the 32-bit stream.

A subnet mask is used to determine the network and host portions.



IPv4 Address Structure

The Subnet Mask

To identify the network and host portions of an IPv4 address, the subnet mask is compared to the IPv4 address bit for bit, from left to right.

- The actual process used to identify the network and host portions is called ANDing.

	Network Portion						Host Portion
IPv4 Address	192 . 168 . 10 .						10
	11000000 10101000 00001010						00001010
Subnet Mask	255 . 255 . 255 .						0
	11111111 11111111 11111111						00000000

IPv4 Address Structure

The Prefix Length

A prefix length is a less cumbersome method used to identify a subnet mask address.

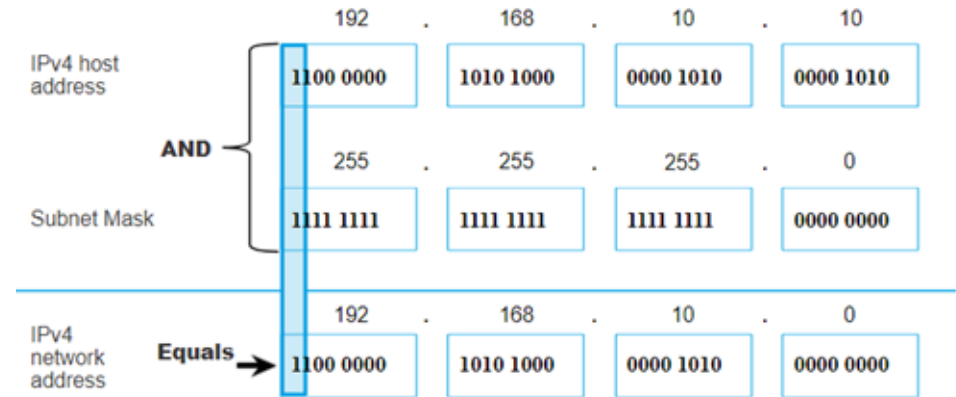
- The prefix length is the number of bits set to 1 in the subnet mask.
- It is written in “slash notation” therefore, count the number of bits in the subnet mask and prepend it with a slash.

Subnet Mask	32-bit Address	Prefix Length
255.0.0.0	11111111.00000000.00000000.00000000	/8
255.255.0.0	11111111.11111111.00000000.00000000	/16
255.255.255.0	11111111.11111111.11111111.00000000	/24
255.255.255.128	11111111.11111111.11111111.10000000	/25
255.255.255.192	11111111.11111111.11111111.11000000	/26
255.255.255.224	11111111.11111111.11111111.11100000	/27
255.255.255.240	11111111.11111111.11111111.11110000	/28
255.255.255.248	11111111.11111111.11111111.11111000	/29
255.255.255.252	11111111.11111111.11111111.11111100	/30

Determining the Network: Logical AND

A logical AND Boolean operation is used in determining the network address.

- Logical AND is the comparison of two bits where only a 1 AND 1 produces a 1 and any other combination results in a 0.
- $1 \text{ AND } 1 = 1$, $0 \text{ AND } 1 = 0$, $1 \text{ AND } 0 = 0$, $0 \text{ AND } 0 = 0$
- $1 = \text{True}$ and $0 = \text{False}$
- To identify the network address, the host IPv4 address is logically ANDed, bit by bit, with the subnet mask to identify the network address.

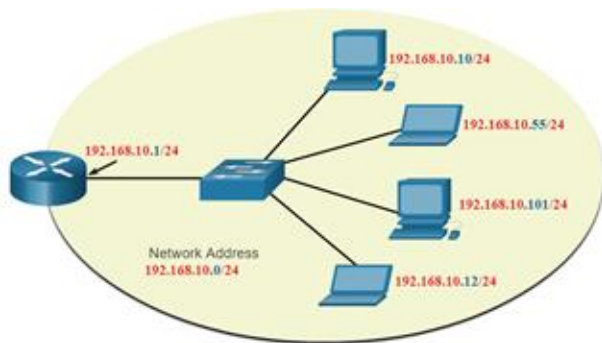


IPv4 Address Structure

Network, Host, and Broadcast Addresses

Within each network are three types of IP addresses:

- Network address
- Host addresses
- Broadcast address



	Network Portion			Host Portion	Host Bits
Subnet mask 255.255.255.0 or /24	255 11111111	255 11111111	255 11111111	0 00000000	
Network address 192.168.10.0 or /24	192 11000000	168 10100000	10 00001010	0 00000000	All 0s
First address 192.168.10.1 or /24	192 11000000	168 10100000	10 00001010	1 00000001	All 0s and a 1
Last address 192.168.10.254 or /24	192 11000000	168 10100000	10 00001010	254 11111110	All 1s and a 0
Broadcast address 192.168.10.255 or /24	192 11000000	168 10100000	10 00001010	255 11111111	All 1s

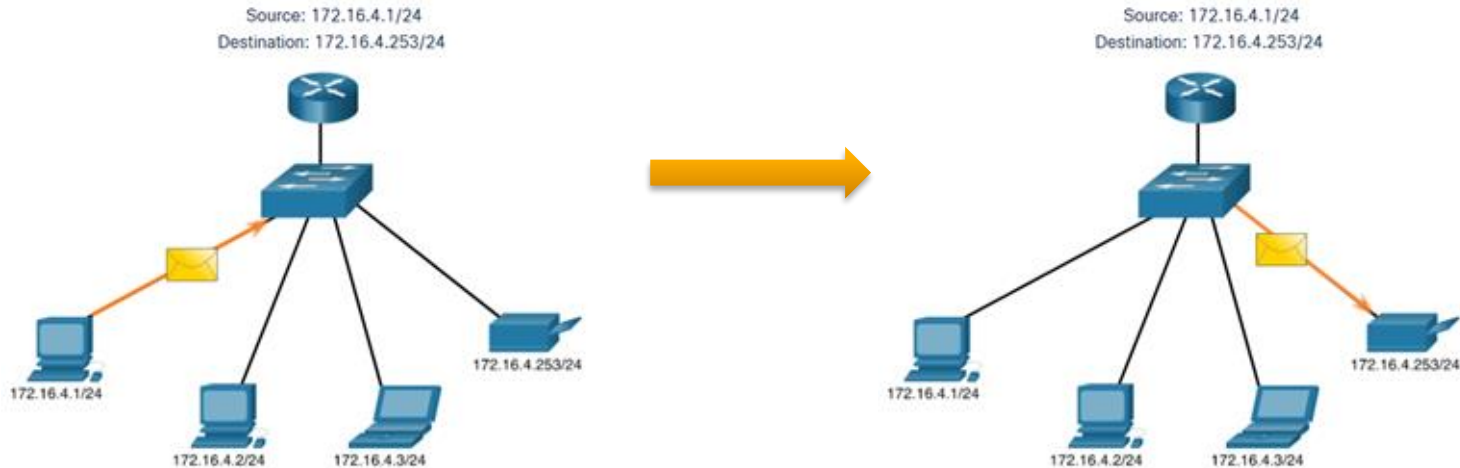
11.2 IPv4 Unicast, Broadcast, and Multicast

IPv4 Unicast, Broadcast, and Multicast

Unicast

Unicast transmission is sending a packet to one destination IP address.

For example, the PC at 172.16.4.1 sends a unicast packet to the printer at 172.16.4.253.



IPv4 Unicast, Broadcast, and Multicast

Broadcast

Broadcast transmission is sending a packet to all other destination IP addresses.

For example, the PC at 172.16.4.1 sends a broadcast packet to all IPv4 hosts.



IPv4 Unicast, Broadcast, and Multicast

Multicast

Multicast transmission is sending a packet to a multicast address group.

For example, the PC at 172.16.4.1 sends a multicast packet to the multicast group address 224.10.10.5.



11.3 Types of IPv4 Addresses

Special Use IPv4 Addresses

Loopback addresses

127.0.0.0 /8 (127.0.0.1 to 127.255.255.254)

Commonly identified as only 127.0.0.1

Used on a host to test if TCP/IP is operational.

```
C:\Users\NetAcad> ping 127.0.0.1
Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
```

Link-Local addresses

- 169.254.0.0 /16 (169.254.0.1 to 169.254.255.254)
- Commonly known as the Automatic Private IP Addressing (APIPA) addresses or self-assigned addresses.
- Used by Windows DHCP clients to self-configure when no DHCP servers are available.

Legacy Classful Addressing

RFC 790 (1981) allocated IPv4 addresses in classes

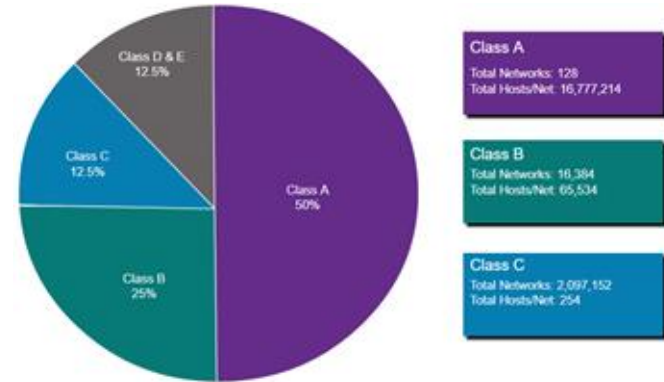
Class A (0.0.0.0/8 to 127.0.0.0/8)

Class B (128.0.0.0 /16 – 191.255.0.0 /16)

Class C (192.0.0.0 /24 – 223.255.255.0 /24)

Class D (224.0.0.0 to 239.0.0.0)

Class E (240.0.0.0 – 255.0.0.0)



Classful addressing wasted many IPv4 addresses.

Classful address allocation was replaced with classless addressing which ignores the rules of classes (A, B, C).

Legacy Classful Addressing

A: 0bbb bbbb. B. B. B >

Legacy Classful Addressing

A: 0bbb bbbb. B. B. B > [0.0.0.0 – 127.255.255.255]

Legacy Classful Addressing

A: 0bbb bbbb. B. B. B > [0.0.0.0 – 127.255.255.255]

B: 10bb bbbb. B. B. B >

Types of IPv4 Addresses

Legacy Classful Addressing

A: 0bbb bbbb. B. B. B > [0.0.0.0 – 127.255.255.255]

B: 10bb bbbb. B. B. B > [128.0.0.0 – 191.255.255.255]

Legacy Classful Addressing

A: 0bbb bbbb. B. B. B > [0.0.0.0 – 127.255.255.255]

B: 10bb bbbb. B. B. B > [128.0.0.0 – 191.255.255.255]

C: 110b bbbb. B. B. B >

Legacy Classful Addressing

A: 0bbb bbbb. B. B. B > [0.0.0.0 – 127.255.255.255]

B: 10bb bbbb. B. B. B > [128.0.0.0 – 191.255.255.255]

C: 110b bbbb. B. B. B > [192.0.0.0 – 223.255.255.255]

Legacy Classful Addressing

A: 0bbb bbbb. B. B. B > [0.0.0.0 – 127.255.255.255]

B: 10bb bbbb. B. B. B > [128.0.0.0 – 191.255.255.255]

C: 110b bbbb. B. B. B > [192.0.0.0 – 223.255.255.255]

D: 1110 bbbb. B. B. B >

Legacy Classful Addressing

A: 0bbb bbbb. B. B. B > [0.0.0.0 – 127.255.255.255]

B: 10bb bbbb. B. B. B > [128.0.0.0 – 191.255.255.255]

C: 110b bbbb. B. B. B > [192.0.0.0 – 223.255.255.255]

D: 1110 bbbb. B. B. B > [224.0.0.0 – 239.255.255.255]

Legacy Classful Addressing

A: 0bbb bbbb. B. B. B > [0.0.0.0 – 127.255.255.255]

B: 10bb bbbb. B. B. B > [128.0.0.0 – 191.255.255.255]

C: 110b bbbb. B. B. B > [192.0.0.0 – 223.255.255.255]

D: 1110 bbbb. B. B. B > [224.0.0.0 – 239.255.255.255]

E: 1111 bbbb. B. B. B >

Legacy Classful Addressing

A: 0bbb bbbb. B. B. B > [0.0.0.0 – 127.255.255.255]

B: 10bb bbbb. B. B. B > [128.0.0.0 – 191.255.255.255]

C: 110b bbbb. B. B. B > [192.0.0.0 – 223.255.255.255]

D: 1110 bbbb. B. B. B > [224.0.0.0 – 239.255.255.255]

E: 1111 bbbb. B. B. B > [240.0.0.0 – 255.255.255.255]

Legacy Classful Addressing

A: 0bbb bbbb. B. B. B /8 = 255.0.0.0
B: 10bb bbbb. B. B. B /16 = 255.255.0.0
C: 110b bbbb. B. B. B /24 = 255.255.255.0

A: 2^7-2 retele si $2^{24}-2$ hosturi/retea
B: 2^{14} retele si $2^{16}-2$ hosturi/retea
C: 2^{21} retele si 2^8-2 hosturi/retea

A: 0bbb bbbb. B. B. B > [0.0.0.0 – 127.255.255.255]
B: 10bb bbbb. B. B. B > [128.0.0.0 – 191.255.255.255]
C: 110b bbbb. B. B. B > [192.0.0.0 – 223.255.255.255]

Types of IPv4 Addresses

Public and Private IPv4 Addresses

As defined in in RFC 1918, public IPv4 addresses are globally routed between internet service provider (ISP) routers.

- Private addresses are common blocks of addresses used by most organizations to assign IPv4 addresses to internal hosts.
- Private IPv4 addresses are not unique and can be used internally within any network.

Network Address and Prefix	RFC 1918 Private Address Range
10.0.0.0/8	10.0.0.0 - 10.255.255.255
172.16.0.0/12	172.16.0.0 - 172.31.255.255
192.168.0.0/16	192.168.0.0 - 192.168.255.255

However, private addresses are not globally routable.

Types of IPv4 Addresses

Routing to the Internet

Private address blocks are:

- 10.0.0.0 to 10.255.255.255 (10.0.0.0/8)
- 172.16.0.0 to 172.31.255.255 (172.16.0.0/12)
- 192.168.0.0 to 192.168.255.255 (192.168.0.0/16)

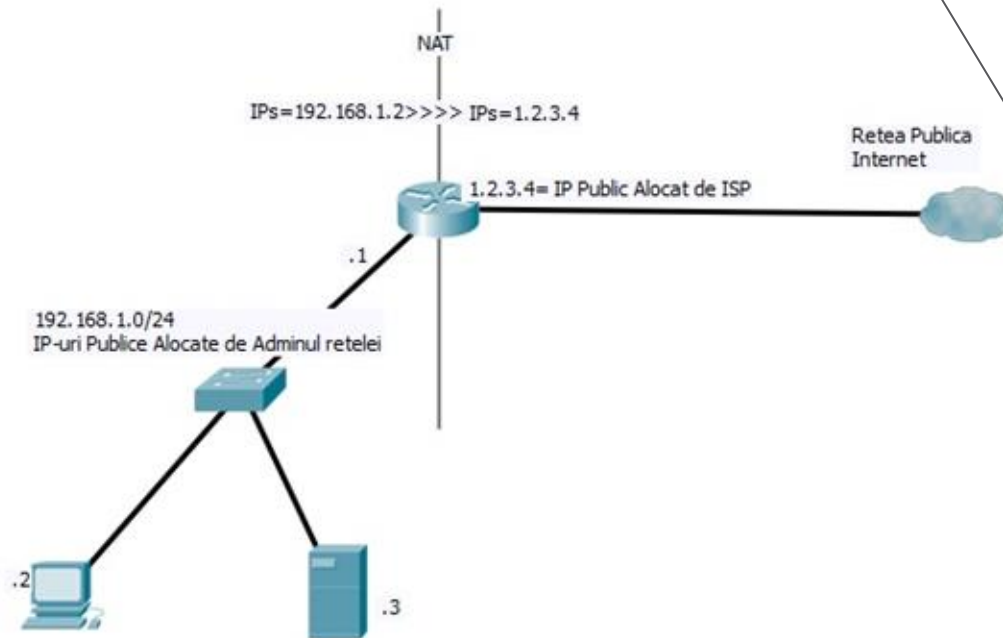
172.16.0.0/16
172.17.0.0/16
172.18.0.0/16

...
172.30.0.0/16
172.31.0.0/16

192.168.0.0/24
192.168.1.0/24
192.168.2.0/24

...
192.168.254.0/24
192.168.255.0/24

NAT

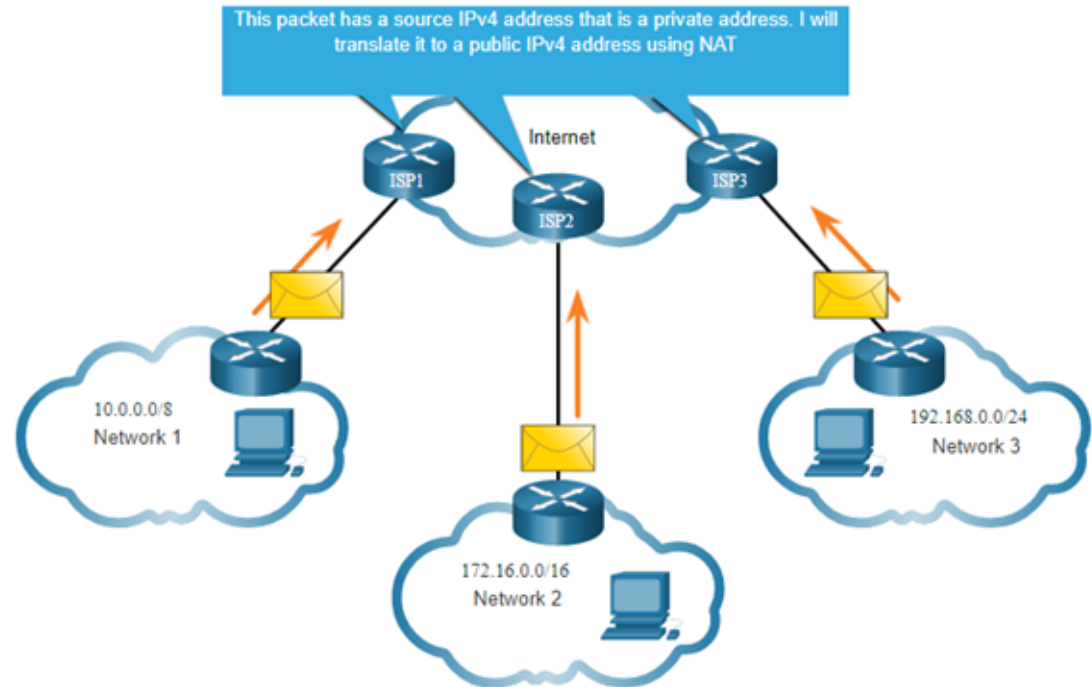


Types of IPv4 Addresses

Routing to the Internet

Network Address Translation (NAT) translates private IPv4 addresses to public IPv4 addresses.

- NAT is typically enabled on the edge router connecting to the internet.
- It translates the internal private address to a public global IP address.



Types of IPv4 Addresses

Assignment of IP Addresses

The Internet Assigned Numbers Authority (IANA) manages and allocates blocks of IPv4 and IPv6 addresses to five Regional Internet Registries (RIRs).

- RIRs are responsible for allocating IP addresses to ISPs who provide IPv4 address blocks to smaller ISPs and organizations.

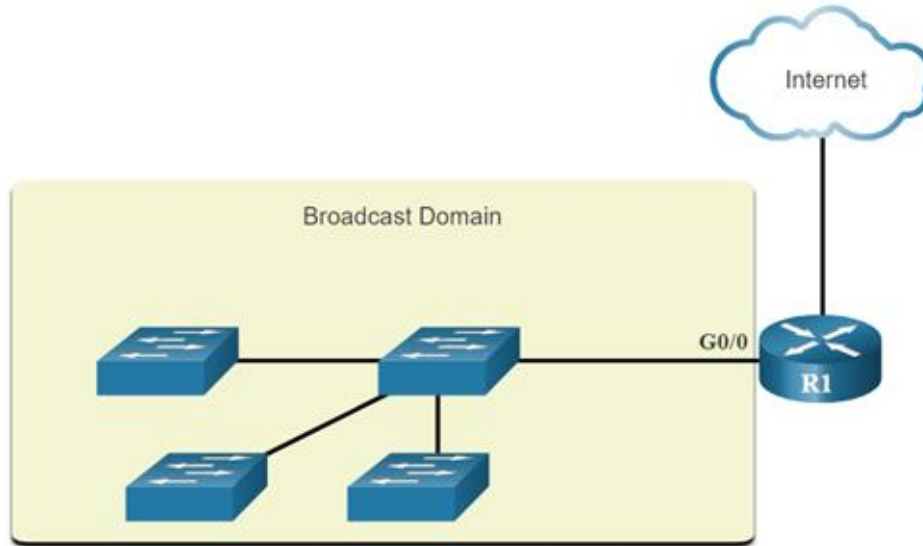


11.4 Network Segmentation

Broadcast Domains and Segmentation

Many protocols use broadcasts or multicasts (e.g., ARP use broadcasts to locate other devices, hosts send DHCP discover broadcasts to locate a DHCP server.)

Switches propagate broadcasts out all interfaces except the interface on which it was received.



- The only device that stops broadcasts is a router.
- Routers do not propagate broadcasts.
- Each router interface connects to a broadcast domain and broadcasts are only propagated within that specific broadcast domain.

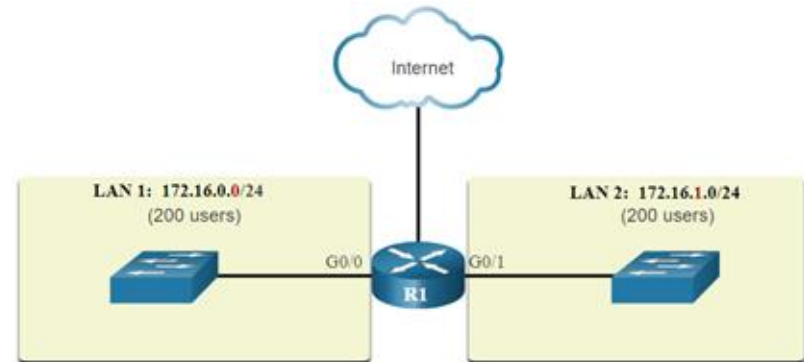
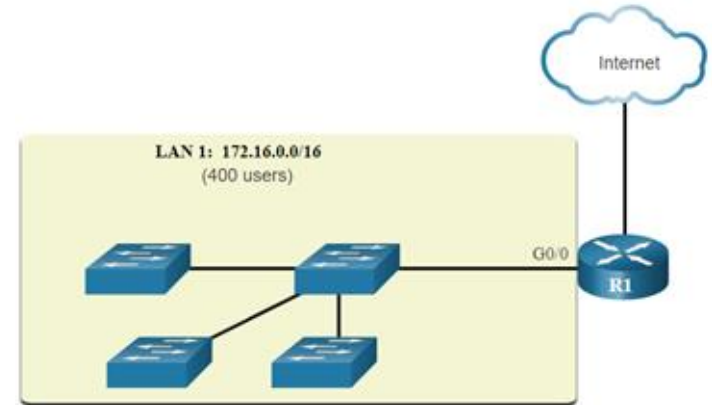
Problems with Large Broadcast Domains

A problem with a large broadcast domain is that these hosts can generate excessive broadcasts and negatively affect the network.

The solution is to reduce the size of the network to create smaller broadcast domains in a process called subnetting.

Dividing the network address 172.16.0.0 /16 into two subnets of 200 users each: 172.16.0.0 /24 and 172.16.1.0 /24.

Broadcasts are only propagated within the smaller broadcast domains.



Network Segmentation

Reasons for Segmenting Networks

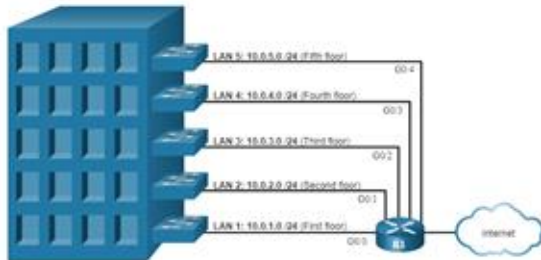
Subnetting reduces overall network traffic and improves network performance.

It can be used to implement security policies between subnets.

Subnetting reduces the number of devices affected by abnormal broadcast traffic.

Subnets are used for a variety of reasons including by:

Location



Group or Function



Device Type



11.5 Subnet an IPv4 Network

Subnet on an Octet Boundary

Networks are most easily subnetted at the octet boundary of /8, /16, and /24.

Notice that using longer prefix lengths decreases the number of hosts per subnet.

Prefix Length	Subnet Mask	Subnet Mask in Binary (n = network, h = host)	# of hosts
/8	255.0.0.0	<code>nnnnnnnnn . hhhhhhhh . hhhhhhhh . hhhhhhhh</code> <code>11111111 . 00000000 . 00000000 . 00000000</code>	16,777,214
/16	255.255.0.0	<code>nnnnnnnnn . nnnnnnnnn . hhhhhhhh . hhhhhhhh</code> <code>11111111 . 11111111 . 00000000 . 00000000</code>	65,534
/24	255.255.255.0	<code>nnnnnnnnn . nnnnnnnnn . nnnnnnnnn . hhhhhhhh</code> <code>11111111 . 11111111 . 11111111 . 00000000</code>	254

Subnet on an Octet Boundary (Cont.)

In the first table 10.0.0.0/8 is subnetted using /16 and in the second table, a /24 mask.

Subnet Address (256 Possible Subnets)	Host Range (65,534 possible hosts per subnet)	Broadcast
10.0.0.0/16	10.0.0.1 - 10.0.255.254	10.0.255.255
10.1.0.0/16	10.1.0.1 - 10.1.255.254	10.1.255.255
10.2.0.0/16	10.2.0.1 - 10.2.255.254	10.2.255.255
10.3.0.0/16	10.3.0.1 - 10.3.255.254	10.3.255.255
10.4.0.0/16	10.4.0.1 - 10.4.255.254	10.4.255.255
10.5.0.0/16	10.5.0.1 - 10.5.255.254	10.5.255.255
10.6.0.0/16	10.6.0.1 - 10.6.255.254	10.6.255.255
10.7.0.0/16	10.7.0.1 - 10.7.255.254	10.7.255.255
...	...	...
10.255.0.0/16	10.255.0.1 - 10.255.255.254	10.255.255.255

Subnet Address (65,536 Possible Subnets)	Host Range (254 possible hosts per subnet)	Broadcast
10.0.0.0/24	10.0.0.1 - 10.0.0.254	10.0.0.255
10.0.1.0/24	10.0.1.1 - 10.0.1.254	10.0.1.255
10.0.2.0/24	10.0.2.1 - 10.0.2.254	10.0.2.255
...	...	...
10.0.255.0/24	10.0.255.1 - 10.0.255.254	10.0.255.255
10.1.0.0/24	10.1.0.1 - 10.1.0.254	10.1.0.255
10.1.1.0/24	10.1.1.1 - 10.1.1.254	10.1.1.255
10.1.2.0/24	10.1.2.1 - 10.1.2.254	10.1.2.255
...	...	...
10.100.0.0/24	10.100.0.1 - 10.100.0.254	10.100.0.255
...	...	...
10.255.255.0/24	10.255.255.1 - 10.255.255.254	10.255.255.255

Subnet within an Octet Boundary

Refer to the table to see six ways to subnet a /24 network.

Prefix Length	Subnet Mask	Subnet Mask in Binary (n = network, h = host)	# of subnets	# of hosts
/25	255.255.255.128	nnnnnnnn . nnnnnnnn . nnnnnnnn . nhhhhhhh 11111111 . 11111111 . 11111111 . 10000000	2	126
/26	255.255.255.192	nnnnnnnn . nnnnnnnn . nnnnnnnn . nnhhhhhh 11111111 . 11111111 . 11111111 . 11000000	4	62
/27	255.255.255.224	nnnnnnnn . nnnnnnnn . nnnnnnnn . nnnhhhhh 11111111 . 11111111 . 11111111 . 11100000	8	30
/28	255.255.255.240	nnnnnnnn . nnnnnnnn . nnnnnnnn . nnnnnhhh 11111111 . 11111111 . 11111111 . 11110000	16	14
/29	255.255.255.248	nnnnnnnn . nnnnnnnn . nnnnnnnn . nnnnnnhh 11111111 . 11111111 . 11111111 . 11111000	32	6
/30	255.255.255.252	nnnnnnnn . nnnnnnnn . nnnnnnnn . nnnnnnhh 11111111 . 11111111 . 11111111 . 11111100	64	2

11.6 Subnet a Slash 16 and a Slash 8 Prefix

Subnet a Slash 16 and a Slash 8 Prefix

Create Subnets with a Slash 16 prefix

The table highlights all the possible scenarios for subnetting a /16 prefix.

Prefix Length	Subnet Mask	Network Address (n = network, h = host)	# of subnets	# of hosts
/17	255.255.128.0	nnnnnnnn.nnnnnnnn.nh h h h h h h h . h h h h h h h h 11111111.11111111.10000000.00000000	2	32766
/18	255.255.192.0	nnnnnnnn.nnnnnnnn.nnh h h h h h h h . h h h h h h h h 11111111.11111111.11000000.00000000	4	16382
/19	255.255.224.0	nnnnnnnn.nnnnnnnn.nnn h h h h h h h . h h h h h h h h 11111111.11111111.11100000.00000000	8	8190
/20	255.255.240.0	nnnnnnnn.nnnnnnnn.nnnn h h h h h . h h h h h h h h 11111111.11111111.11110000.00000000	16	4094
/21	255.255.248.0	nnnnnnnn.nnnnnnnn.nnnnn h h h h . h h h h h h h h 11111111.11111111.11111000.00000000	32	2046
/22	255.255.252.0	nnnnnnnn.nnnnnnnn.nnnnnn h h h . h h h h h h h h 11111111.11111111.11111100.00000000	64	1022
/23	255.255.254.0	nnnnnnnn.nnnnnnnn.nnnnnnn h . h h h h h h h h 11111111.11111111.11111110.00000000	128	510
/24	255.255.255.0	nnnnnnnn.nnnnnnnn.nnnnnnnn . h h h h h h h h 11111111.11111111.11111111.00000000	256	254
/25	255.255.255.128	nnnnnnnn.nnnnnnnn.nnnnnnnn.nh h h h h h h h 11111111.11111111.11111111.10000000	512	126
/26	255.255.255.192	nnnnnnnn.nnnnnnnn.nnnnnnnn.nnh h h h h h h h 11111111.11111111.11111111.11000000	1024	62
/27	255.255.255.224	nnnnnnnn.nnnnnnnn.nnnnnnnn.nnn h h h h h h h 11111111.11111111.11111111.11100000	2048	30
/28	255.255.255.240	nnnnnnnn.nnnnnnnn.nnnnnnnn.nnnn h h h h h 11111111.11111111.11111111.11110000	4096	14
/29	255.255.255.248	nnnnnnnn.nnnnnnnn.nnnnnnnn.nnnnn h h h h 11111111.11111111.11111111.11111000	8192	6
/30	255.255.255.252	nnnnnnnn.nnnnnnnn.nnnnnnnn.nnnnnn h h 11111111.11111111.11111111.11111100	16384	2

Subnet a Slash 16 and a Slash 8 Prefix

Create 100 Subnets with a Slash 16 prefix

Consider a large enterprise that requires at least 100 subnets and has chosen the private address 172.16.0.0/16 as its internal network address.

The figure displays the number of subnets that can be created when borrowing bits from the third octet and the fourth octet.

Notice there are now up to 14 host bits that can be borrowed (i.e., last two bits cannot be borrowed).

To satisfy the requirement of 100 subnets for the enterprise, 7 bits (i.e., $2^7 = 128$ subnets) would need to be borrowed (for a total of 128 subnets).



Subnet a Slash 16 and a Slash 8 Prefix

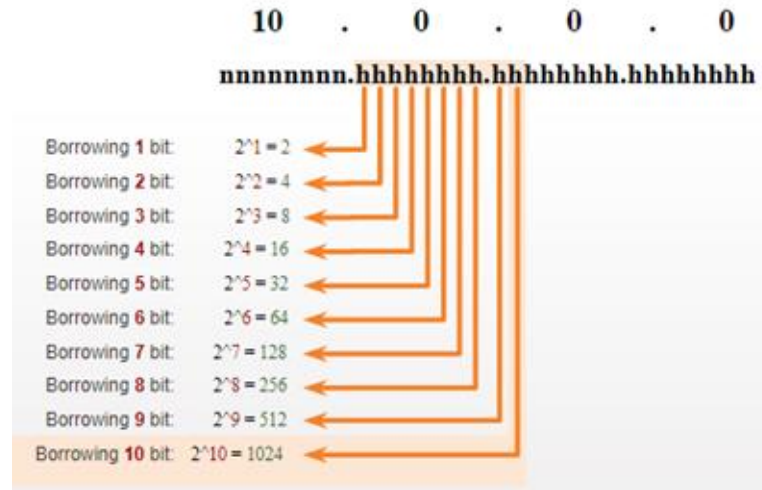
Create 1000 Subnets with a Slash 8 prefix

Consider a small ISP that requires 1000 subnets for its clients using network address 10.0.0.0/8 which means there are 8 bits in the network portion and 24 host bits available to borrow toward subnetting.

The figure displays the number of subnets that can be created when borrowing bits from the second and third.

Notice there are now up to 22 host bits that can be borrowed (i.e., last two bits cannot be borrowed).

To satisfy the requirement of 1000 subnets for the enterprise, 10 bits (i.e., $2^{10}=1024$ subnets) would need to be borrowed (for a total of 128 subnets)



11.7 Subnet to Meet Requirements

Subnet Private versus Public IPv4 Address Space

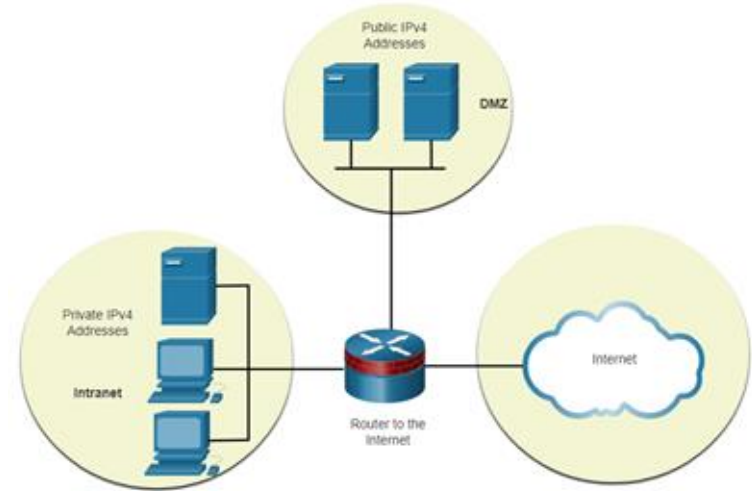
Enterprise networks will have an:

Intranet - A company's internal network typically using private IPv4 addresses.

DMZ – A companies internet facing servers.
Devices in the DMZ use public IPv4 addresses.

A company could use the 10.0.0.0/8 and subnet on the /16 or /24 network boundary.

The DMZ devices would have to be configured with public IP addresses.



Subnet to Meet Requirements

Minimize Unused Host IPv4 Addresses and Maximize Subnets

There are two considerations when planning subnets:

The number of host addresses required for each network

The number of individual subnets needed



Prefix Length	Subnet Mask	Subnet Mask in Binary (n = network, h = host)	# of subnets	# of hosts
/25	255.255.255.128	nnnnnnnn . nnnnnnnn . nnnnnnnn . nhhhhhhh 11111111 . 11111111 . 11111111 . 10000000	2	126
/26	255.255.255.192	nnnnnnnn . nnnnnnnn . nnnnnnnn . nnhhhhhh 11111111 . 11111111 . 11111111 . 11000000	4	62
/27	255.255.255.224	nnnnnnnn . nnnnnnnn . nnnnnnnn . nnnhhhhh 11111111 . 11111111 . 11111111 . 11100000	8	30
/28	255.255.255.240	nnnnnnnn . nnnnnnnn . nnnnnnnn . nnnnnhhh 11111111 . 11111111 . 11111111 . 11110000	16	14
/29	255.255.255.248	nnnnnnnn . nnnnnnnn . nnnnnnnn . nnnnnnhh 11111111 . 11111111 . 11111111 . 11111000	32	6
/30	255.255.255.252	nnnnnnnn . nnnnnnnn . nnnnnnnn . nnnnnnhh 11111111 . 11111111 . 11111111 . 11111100	64	2

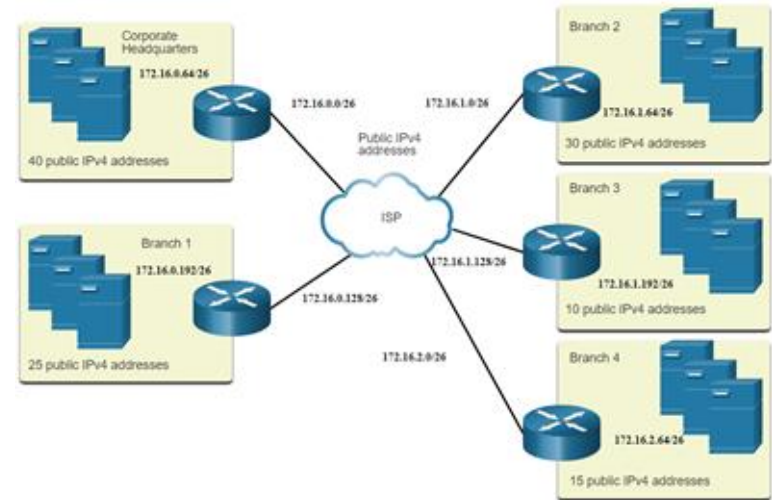
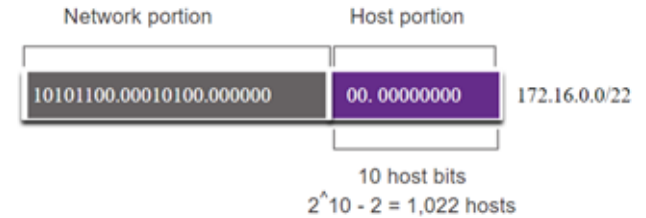
Subnet to Meet Requirements

Example: Efficient IPv4 Subnetting

In this example, corporate headquarters has been allocated a public network address of 172.16.0.0/22 (10 host bits) by its ISP providing 1,022 host addresses.

There are five sites and therefore five internet connections which means the organization requires 10 subnets with the largest subnet requires 40 addresses.

It allocated 10 subnets with a /26 (i.e., 255.255.255.192) subnet mask.

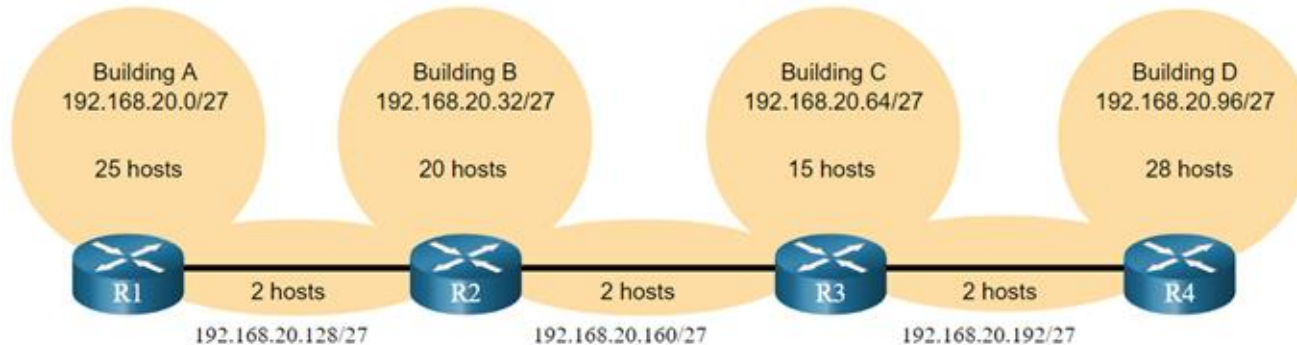


11.8 VLSM

VLSM IPv4 Address Conservation

Given the topology, 7 subnets are required (i.e, four LANs and three WAN links) and the largest number of host is in Building D with 28 hosts.

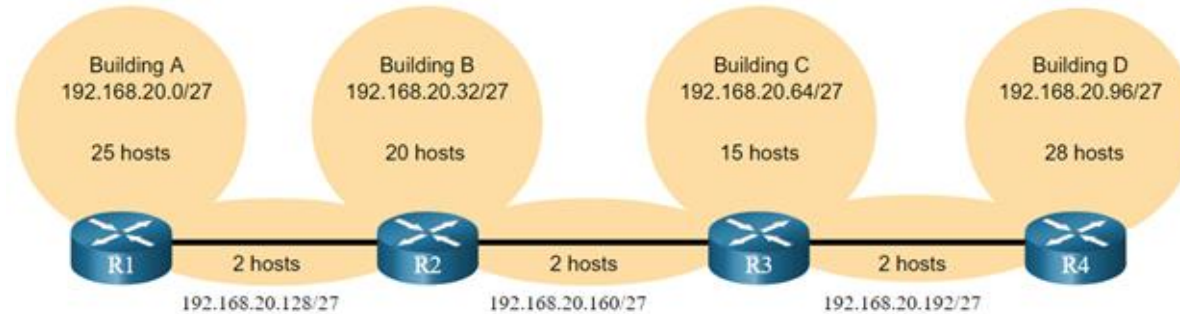
A /27 mask would provide 8 subnets of 30 host IP addresses and therefore support this topology.



IPv4 Address Conservation (Cont.)

However, the point-to-point WAN links only require two addresses and therefore waste 28 addresses each for a total of 84 unused addresses.

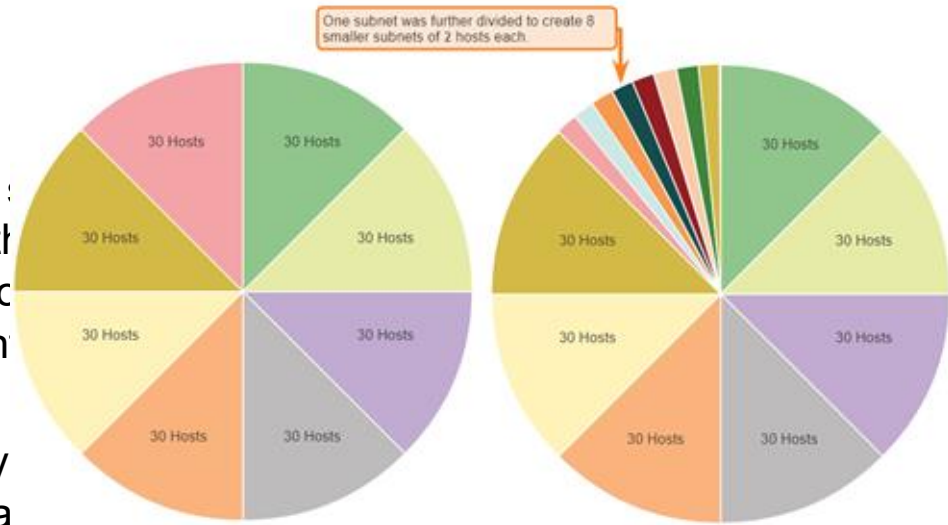
Host portion
 $2^5 - 2 = 30$ host IP addresses per subnet
 $30 - 2 = 28$
Each WAN subnet wastes 28 addresses
 $28 \times 3 = 84$
84 addresses are unused



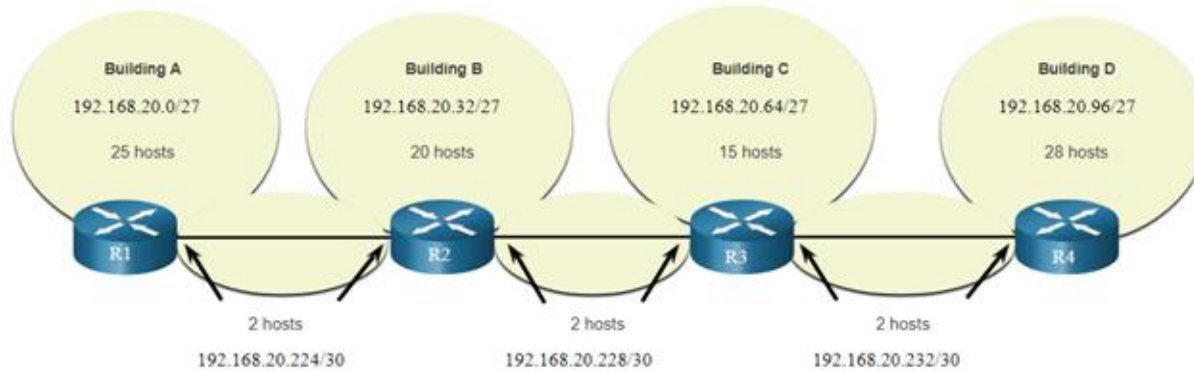
- Applying a traditional subnetting scheme to this scenario is not very efficient and is wasteful.
- VLSM was developed to avoid wasting addresses by enabling us to subnet a subnet.

VLSM VLSM

The left side displays the traditional (i.e., the same subnet mask) while the right side illustrates how VLSM can be used to efficiently divide the last subnet into eight smaller subnets of 2 hosts each.

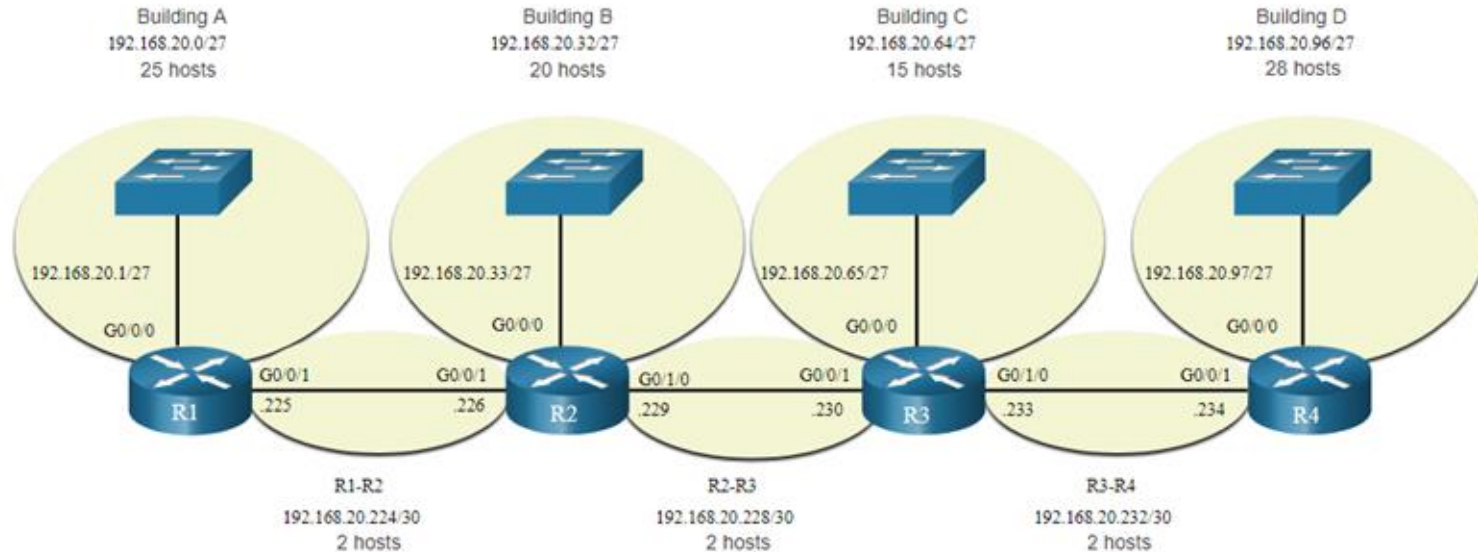


When using VLSM, always begin by identifying the requirements of the largest subnet and then work down to the smallest.

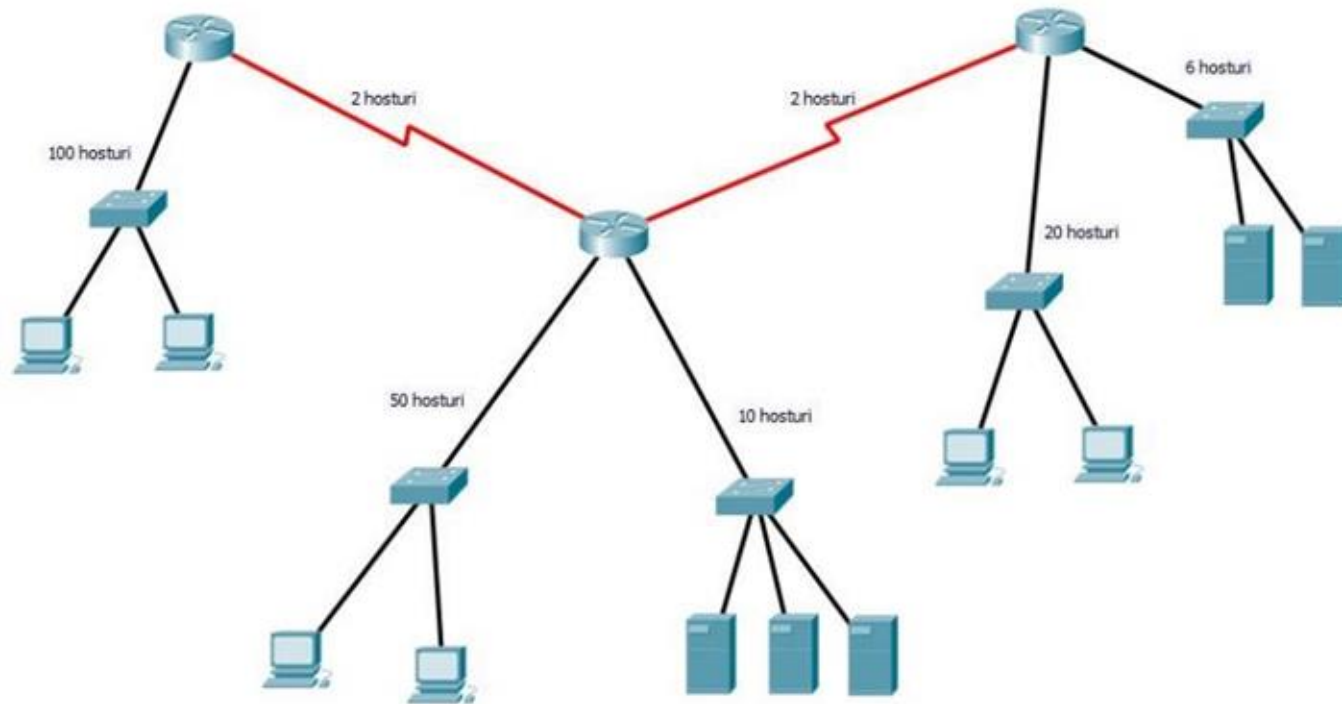


VLSM Topology Address Assignment

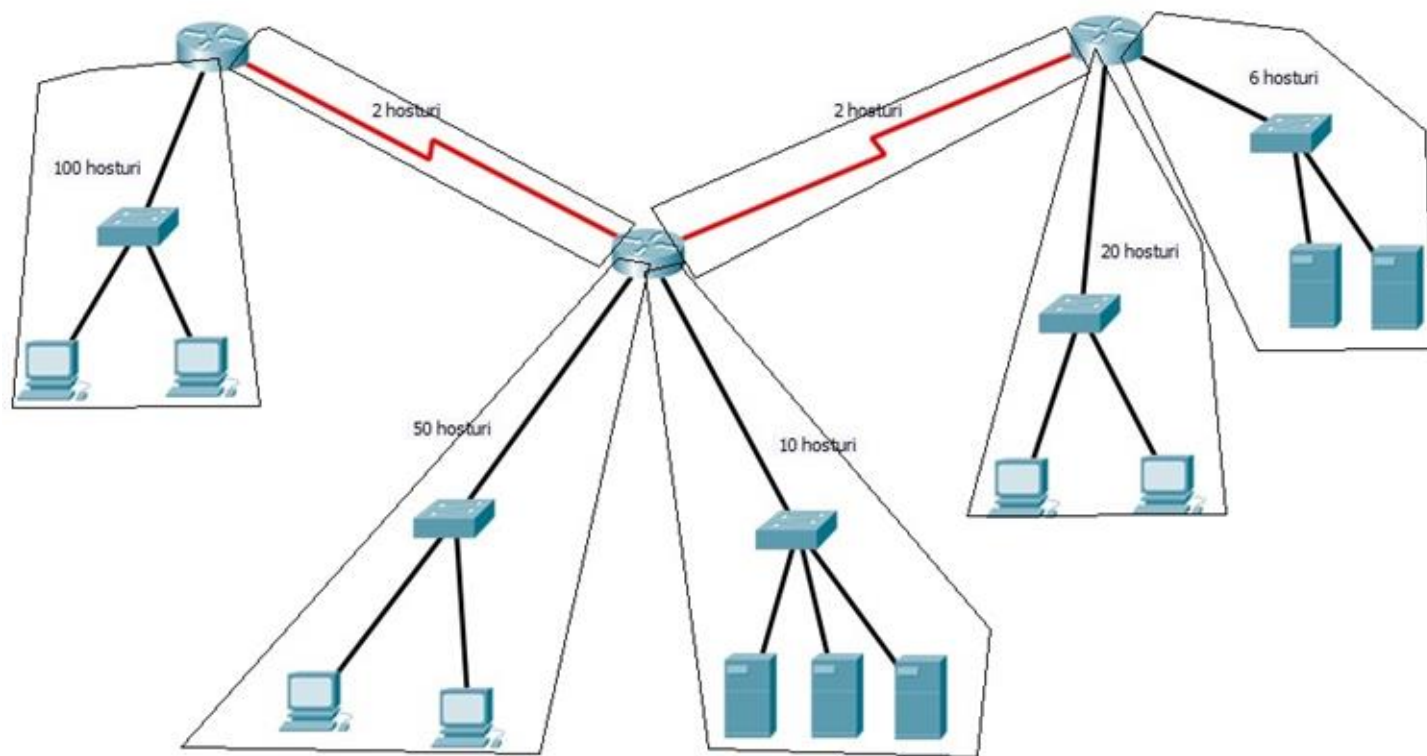
Using VLSM subnets, the LAN and inter-router networks can be addressed without unnecessary waste as shown in the logical topology diagram.



Exercitiul 1 de subnetizare:
Alocați adrese IP cât mai eficient folosind rețeaua 1.0.0.0/24



Domenii de Broadcast și rețele P2P



Hosturi	1.0.0.0/24
100	$100 + 2 \text{ (IP Retea Si IP DB)} \leq 2^7$ - Avem nevoie de 7 biti de host
50	
20	
10	
6	
2	
2	

Hosturi

~~1.0.0.0/24~~

100

$100 + 2 \text{ (IP Retea Si IP DB)} \leq 2^7$ - Avem nevoie de 7 biti de host

50

20

10

6

2

2

Hosturi

~~1.0.0.0/24~~

100

100+ 2 (IP Retea Si IP DB) $\leq 2^7$ - Avem nevoie de 7 biti de host

50

1.0.0.0000 0000

20

10

6

2

2

Hosturi

~~1.0.0.0/24~~

100

100+ 2 (IP Retea Si IP DB) $\leq 2^7$ - Avem nevoie de 7 biti de host

50

1.0.0.0000 0000

20

1

10

6

2

2

Hosturi

~~1.0.0.0/24~~

100

100 + 2 (IP Retea Si IP DB) $\leq 2^7$ - Avem nevoie de 7 biti de host

50

1.0.0.0000 0000

=> 1.0.0.0/25 pentru retea de 100 hosturi

20

1

=> 1.0.0.128/25

10

6

2

2

Hosturi

100

50

20

10

6

2

2

~~1.0.0.0/24~~

100 + 2 (IP Retea Si IP DB) $\leq 2^7$ - Avem nevoie de 7 biti de host

1.0.0.0000 0000 $\Rightarrow$ 1.0.0.0/25 pentru retea de 100 hosturi

1 $\Rightarrow$ 1.0.0.128/25

50 + 2 $\leq 2^6$ - Avem nevoie de 6 biti de host

Hosturi

~~1.0.0.0/24~~

100

100 + 2 (IP Retea Si IP DB) $\leq 2^7$ - Avem nevoie de 7 biti de host

50

1.0.0.0000 0000 $\Rightarrow$ 1.0.0.0/25 pentru retea de 100 hosturi

20

1
 $\Rightarrow$ 1.0.0.128/25

10

6

50 + 2 $\leq 2^6$ - Avem nevoie de 6 biti de host

2

1.0.0.1000 0000

2

Hosturi

~~1.0.0.0/24~~

100

100 + 2 (IP Retea Si IP DB) $\leq 2^7$ - Avem nevoie de 7 biti de host

50

1.0.0.0000 0000 $\Rightarrow$ 1.0.0.0/25 pentru retea de 100 hosturi

20

1

$\Rightarrow$ 1.0.0.128/25

10

6

50 + 2 $\leq 2^6$ - Avem nevoie de 6 biti de host

2

1.0.0.1000 0000

2

1

Hosturi

100

~~1.0.0.0/24~~

$100 + 2$ (IP Retea Si IP DB) $\leq 2^7$ - Avem nevoie de 7 biti de host

50

1.0.0.0000 0000

$\Rightarrow$ 1.0.0.0/25 pentru reteaua de 100 hosturi

20

1

$\Rightarrow$ 1.0.0.128/25

10

6

$50 + 2 \leq 2^6$ - Avem nevoie de 6 biti de host

2

1.0.0.1000 0000

$\Rightarrow$ 1.0.0.128/26 pentru reteaua de 50 hosturi

2

1

$\Rightarrow$ 1.0.0.192/26

Hosturi

100

~~1.0.0.0/24~~

100 + 2 (IP Retea Si IP DB) $\leq 2^7$ - Avem nevoie de 7 biti de host

50

1.0.0.0000 0000 $\Rightarrow$ 1.0.0.0/25 pentru reteaua de 100 hosturi

20

1 $\Rightarrow$ 1.0.0.128/25

10

6

50 + 2 $\leq 2^6$ - Avem nevoie de 6 biti de host

2

1.0.0.1000 0000 $\Rightarrow$ 1.0.0.128/26 pentru reteaua de 50 hosturi

2

1 $\Rightarrow$ 1.0.0.192/26

20 + 2 $\leq 2^5$ - Avem nevoie de 5 biti de host

Hosturi

100

~~1.0.0.0/24~~

100 + 2 (IP Retea Si IP DB) $\leq 2^7$ - Avem nevoie de 7 biti de host

50

1.0.0.0000 0000

$\Rightarrow$ 1.0.0.0/25

pentru reteaua de 100 hosturi

20

1

$\Rightarrow$ 1.0.0.128/25

10

6

50 + 2 $\leq 2^6$ - Avem nevoie de 6 biti de host

2

1.0.0.1000 0000

$\Rightarrow$ 1.0.0.128/26

pentru reteaua de 50 hosturi

2

1

$\Rightarrow$ 1.0.0.192/26

20 + 2 $\leq 2^5$ - Avem nevoie de 5 biti de host

1.0.0.1100 0000

Hosturi

100

~~1.0.0.0/24~~

100 + 2 (IP Retea Si IP DB) $\leq 2^7$ - Avem nevoie de 7 biti de host

50

1.0.0.0000 0000 $\Rightarrow$ 1.0.0.0/25 pentru reteaua de 100 hosturi

20

1 $\Rightarrow$ 1.0.0.128/25

10

6

50 + 2 $\leq 2^6$ - Avem nevoie de 6 biti de host

2

1.0.0.1000 0000 $\Rightarrow$ 1.0.0.128/26 pentru reteaua de 50 hosturi

2

1 $\Rightarrow$ 1.0.0.192/26

20 + 2 $\leq 2^5$ - Avem nevoie de 5 biti de host

1.0.0.1100 0000

1

Hosturi

100

~~1.0.0.0/24~~

100 + 2 (IP Retea Si IP DB) $\leq 2^7$ - Avem nevoie de 7 biti de host

50 1.0.0.0000 0000 $\Rightarrow$ 1.0.0.0/25 pentru reteaua de 100 hosturi

20 1 $\Rightarrow$ 1.0.0.128/25

10

6 50 + 2 $\leq 2^6$ - Avem nevoie de 6 biti de host

2 1.0.0.1000 0000 $\Rightarrow$ 1.0.0.128/26 pentru reteaua de 50 hosturi

2 1 $\Rightarrow$ 1.0.0.192/26

20 + 2 $\leq 2^5$ - Avem nevoie de 5 biti de host

1.0.0.1100 0000 $\Rightarrow$ 1.0.0.192/27 pentru reteaua de 20 hosturi

$\Rightarrow$ 1.0.0.224/27

Hosturi

100

~~1.0.0.0/24~~

100 + 2 (IP Retea Si IP DB) $\leq 2^7$ - Avem nevoie de 7 biti de host

50

1.0.0.0000 0000 $\Rightarrow$ 1.0.0.0/25 pentru reteaua de 100 hosturi

20

1 $\Rightarrow$ 1.0.0.128/25

10

6

50 + 2 $\leq 2^6$ - Avem nevoie de 6 biti de host

2

1.0.0.1000 0000 $\Rightarrow$ 1.0.0.128/26 pentru reteaua de 50 hosturi

2

1 $\Rightarrow$ 1.0.0.192/26

20 + 2 $\leq 2^5$ - Avem nevoie de 5 biti de host

1.0.0.1100 0000 $\Rightarrow$ 1.0.0.192/27 pentru reteaua de 20 hosturi

1 $\Rightarrow$ 1.0.0.224/27

10 + 2 $\leq 2^4$ - Avem nevoie de 4 biti de host

Hosturi

100

~~1.0.0.0/24~~

100 + 2 (IP Retea Si IP DB) $\leq 2^7$ - Avem nevoie de 7 biti de host

50

1.0.0.0000 0000 $\Rightarrow$ 1.0.0.0/25 pentru reteaua de 100 hosturi

20

1 $\Rightarrow$ 1.0.0.128/25

10

6

50 + 2 $\leq 2^6$ - Avem nevoie de 6 biti de host

2

1.0.0.1000 0000 $\Rightarrow$ 1.0.0.128/26 pentru reteaua de 50 hosturi

2

1 $\Rightarrow$ 1.0.0.192/26

20 + 2 $\leq 2^5$ - Avem nevoie de 5 biti de host

1.0.0.1100 0000 $\Rightarrow$ 1.0.0.192/27 pentru reteaua de 20 hosturi

1 $\Rightarrow$ 1.0.0.224/27

10 + 2 $\leq 2^4$ - Avem nevoie de 4 biti de host

1.0.0.1110 0000

Hosturi

100

~~1.0.0.0/24~~

100 + 2 (IP Retea Si IP DB) $\leq 2^7$ - Avem nevoie de 7 biti de host

50

1.0.0.0000 0000 $\Rightarrow$ 1.0.0.0/25 pentru reteaua de 100 hosturi

20

1 $\Rightarrow$ 1.0.0.128/25

10

6

50 + 2 $\leq 2^6$ - Avem nevoie de 6 biti de host

2

1.0.0.1000 0000 $\Rightarrow$ 1.0.0.128/26 pentru reteaua de 50 hosturi

2

1 $\Rightarrow$ 1.0.0.192/26

20 + 2 $\leq 2^5$ - Avem nevoie de 5 biti de host

1.0.0.1100 0000 $\Rightarrow$ 1.0.0.192/27 pentru reteaua de 20 hosturi

1 $\Rightarrow$ 1.0.0.224/27

10 + 2 $\leq 2^4$ - Avem nevoie de 4 biti de host

1.0.0.1110 0000 $\Rightarrow$ 1.0.0.224/28 pentru reteaua de 10 hosturi

$\Rightarrow$ 1.0.0.240/28

Hosturi		$10 + 2 \leq 2^4$ - Avem nevoie de 4 biti de host	
100	1.0.0. 1110 0000	=>	1.0.0.224/28 pentru retea de 10 hosturi
50	1	=>	1.0.0.240/28
20			
10		$6 + 2 \leq 2^3$ - Avem nevoie de 3 biti de host	
6			
2			
2			

Hosturi		$10 + 2 \leq 2^4$ - Avem nevoie de 4 biti de host	
100	1.0.0. 1110 0000	=>	1.0.0.224/28 pentru retea de 10 hosturi
50	1	=>	1.0.0.240/28
20			
10		$6 + 2 \leq 2^3$ - Avem nevoie de 3 biti de host	
6	1.0.0. 1111 0000		
2			
2			

Hosturi			
100	1.0.0. 1110 0000	=>	1.0.0.224/28 pentru retea de 10 hosturi
50	1	=>	1.0.0.240/28
20			
10			
	6 + 2 =< 2^3		- Avem nevoie de 3 biti de host
6	1.0.0. 1111 0000		
2			
2			

Hosturi			
100	1.0.0. 1110 0000	=>	1.0.0.224/28 pentru retea de 10 hosturi
50	1	=>	1.0.0.240/28
20			
10	6 + 2 =< 2^3 - Avem nevoie de 3 biti de host		
6	1.0.0. 1111 0000	=>	1.0.0.240/29 pentru retea de 6 hosturi
2	1	=>	1.0.0.248/29
2			

Hosturi			
100	1.0.0. 1110 0000	=>	1.0.0.224/28 pentru retea de 10 hosturi
50	1	=>	1.0.0.240/28
20			
10	6 + 2 =< 2^3 - Avem nevoie de 3 biti de host		
6	1.0.0. 1111 0000	=>	1.0.0.240/29 pentru retea de 6 hosturi
2	1	=>	1.0.0.248/29
2			
	2 + 2 =< 2^2 - Avem nevoie de 2 biti de host		

Hosturi			
100	1.0.0. 1110 0000	=>	1.0.0.224/28 pentru retea de 10 hosturi
50	1	=>	1.0.0.240/28
20			
10	6 + 2 =< 2^3 - Avem nevoie de 3 biti de host		
6	1.0.0. 1111 0000	=>	1.0.0.240/29 pentru retea de 6 hosturi
2	1	=>	1.0.0.248/29
2			
	2 + 2 =< 2^2 - Avem nevoie de 2 biti de host		
	1.0.0. 1111 1000		

Hosturi

$10 + 2 \leq 2^4$ - Avem nevoie de 4 biti de host

100

1.0.0. 1110 0000 $\Rightarrow$ 1.0.0.224/28 pentru retea de 10 hosturi

50

$\Rightarrow$ 1.0.0.240/28

20

10

$6 + 2 \leq 2^3$ - Avem nevoie de 3 biti de host

6

1.0.0. 1111 0000 $\Rightarrow$ 1.0.0.240/29 pentru retea de 6 hosturi

2

1 $\Rightarrow$ 1.0.0.248/29

2

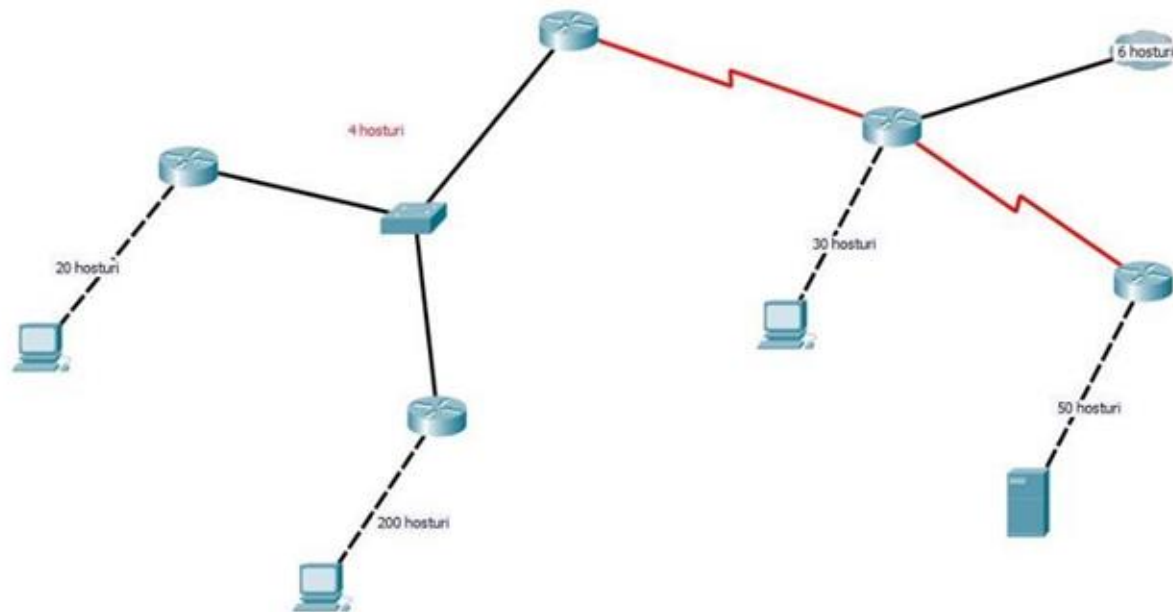
$2 + 2 \leq 2^2$ - Avem nevoie de 2 biti de host

1.0.0. 1111 1000

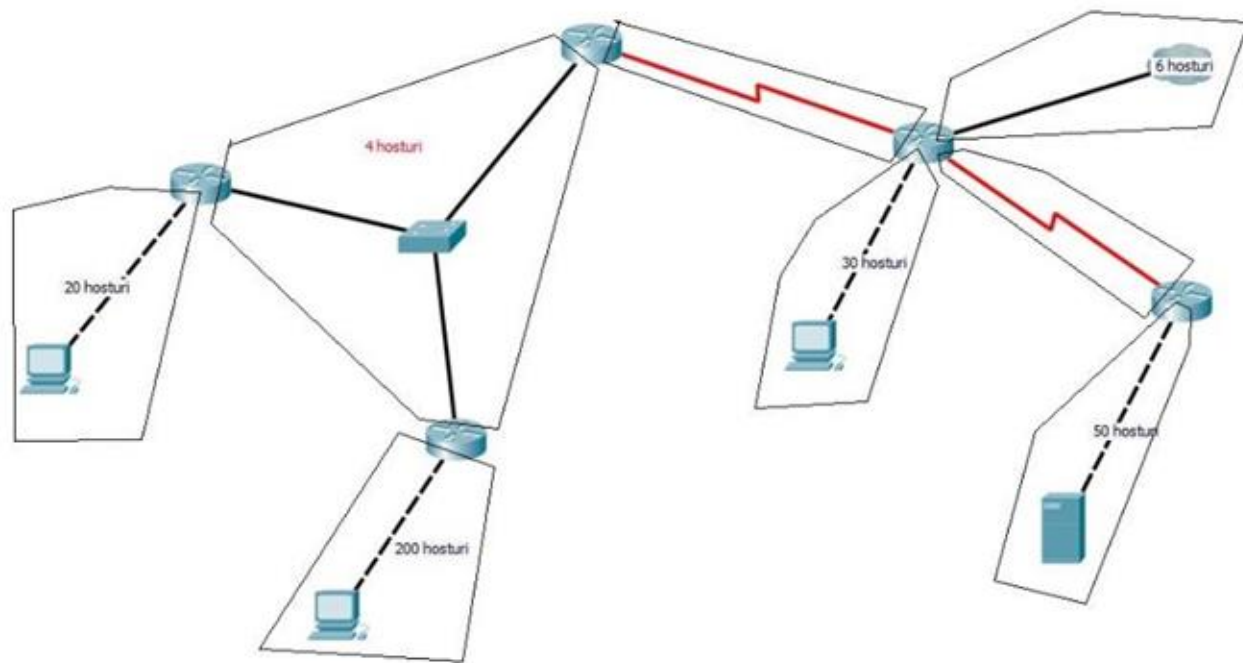
1

Hosturi			
100	1.0.0. 1110 0000	=> 1.0.0.224/28	pentru reteaua de 10 hosturi
50	1	=> 1.0.0.240/28	
20			
10	6 + 2 =< 2^3 - Avem nevoie de 3 biti de host		
6	1.0.0. 1111 0000	=> 1.0.0.240/29	pentru reteaua de 6 hosturi
2	1	=> 1.0.0.248/29	
2			
	2 + 2 =< 2^2 - Avem nevoie de 2 biti de host		
	1.0.0. 1111 1000	=> 1.0.0.248/30	pentru prima retea de 2 hosturi
	1	=> 1.0.0.252/30	a doua retea de 2 hosturi

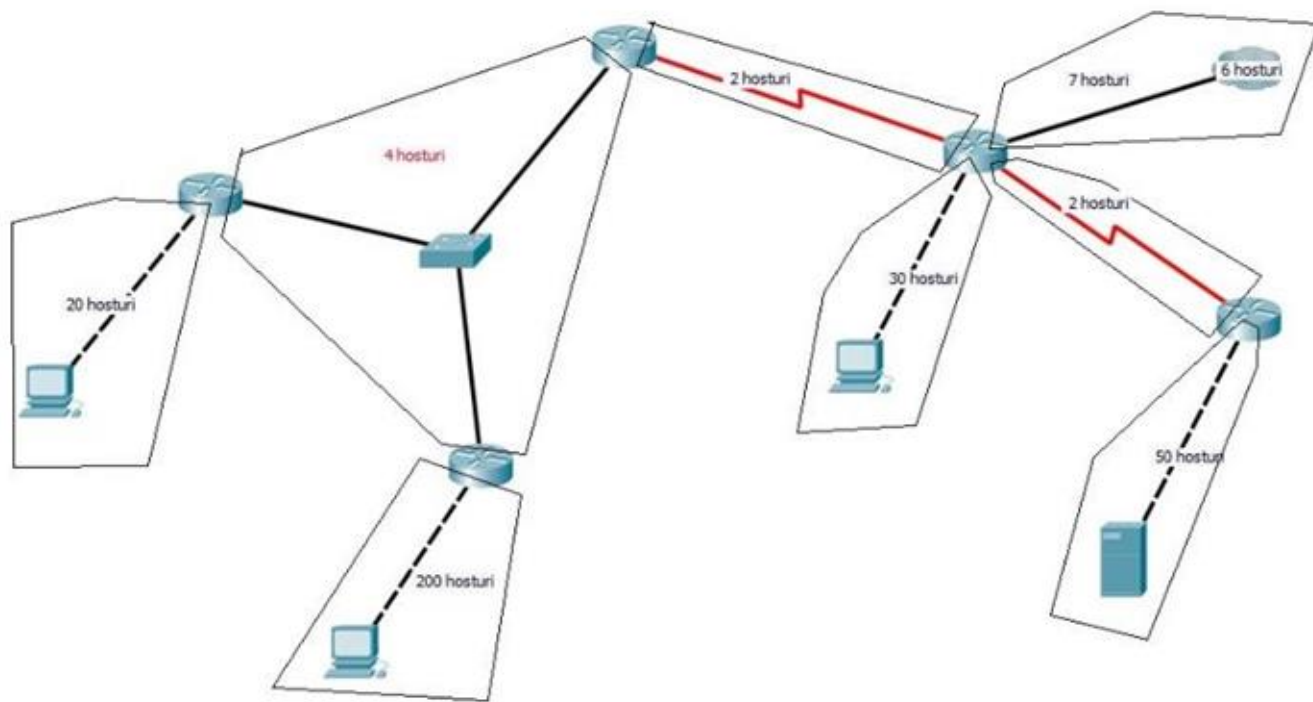
Ex. 2. Subnetizati 2.0.0.0/23



Ex. 2. Subnetizati 2.0.0.0/23



Ex. 2. Subnetizati 2.0.0.0/23



Ex. 2. Subnetizati 2.0.0.0/23

Hosturi

200

50

30

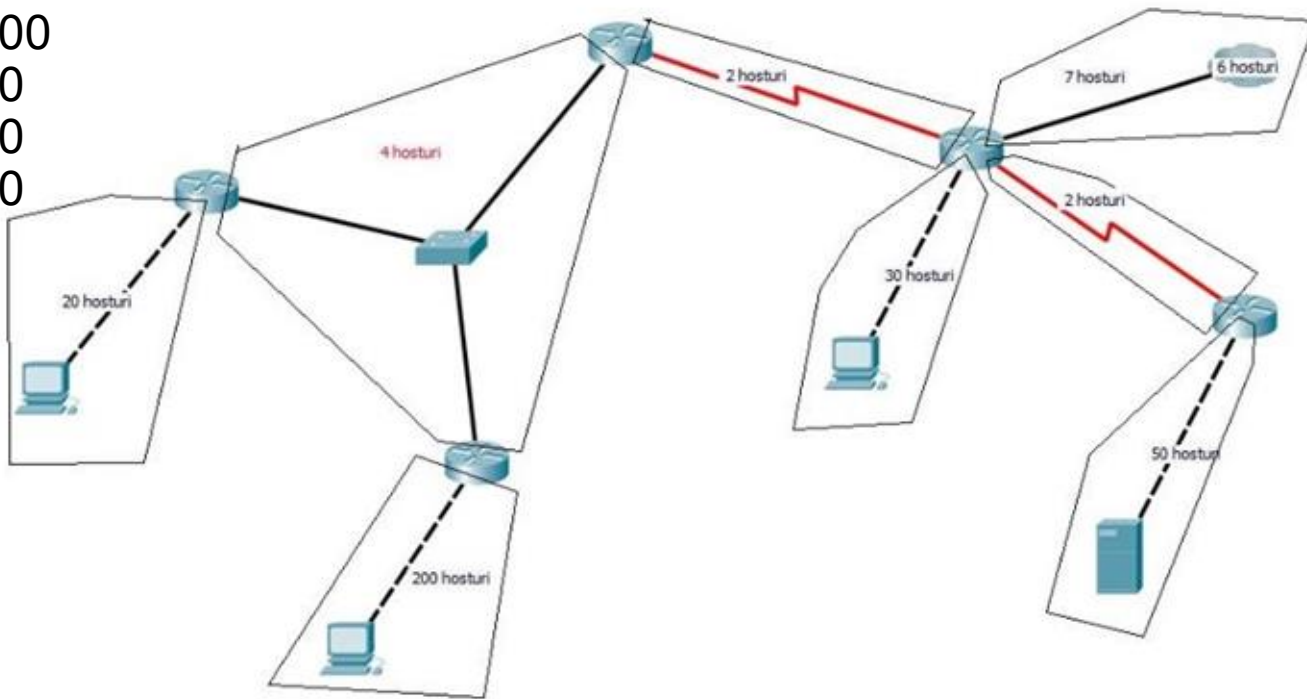
20

7

4

2

2



Hosturi

200
50
30
20
7
4
2
2

2.0.0.0/23

$200+2 \leq 2^8 \Rightarrow$ avem nevoie de 8 biti de host

Hosturi

200

50

30

20

7

4

2

2

2.0.0.0/23

$200+2 \leq 2^8 \Rightarrow$ avem nevoie de 8 biti de host

Hosturi

200
50
30
20
7
4
2
2

2.0.0.0/23

$200+2 \leq 2^8 \Rightarrow$ avem nevoie de 8 biti de host

2.0.0000 0000.0000 0000

Hosturi

200
50
30
20
7
4
2
2

2.0.0.0/23

$200+2 \leq 2^8 \Rightarrow$ avem nevoie de 8 biti de host

2.0.0000 0000 | 0000 0000
 | 1

Hosturi

200

50

30

20

7

4

2

2

2.0.0.0/23

$200+2 \leq 2^8 \Rightarrow$ avem nevoie de 8 biti de host

2.0. 0000 0000 | 0000 0000

2.0.0.0/24

pentru rețeaua de 200 hosturi

2.0.1.0/24

Hosturi

200

50

30

20

7

4

2

2

2.0.0.0/23

$200+2 \leq 2^8 \Rightarrow$ avem nevoie de 8 biti de host

2.0. 0000 0000 | 0000 0000

2.0.0.0/24

pentru rețeaua de 200 hosturi

2.0.1.0/24

1

$50+2 \leq 2^6 \Rightarrow$ avem nevoie de 6 biti de host

Hosturi

200

50

30

20

7

4

2

2

2.0.0.0/23

$200+2 \leq 2^8 \Rightarrow$ avem nevoie de 8 biti de host

2.0. 0000 0000 | 0000 0000

2.0.0.0/24

pentru retea de 200 hosturi

2.0.1.0/24

$50+2 \leq 2^6 \Rightarrow$ avem nevoie de 6 biti de host

2.0.1. 0000 0000

Hosturi

200

50

30

20

7

4

2

2

2.0.0.0/23

$200+2 \leq 2^8 \Rightarrow$ avem nevoie de 8 biti de host

2.0. 0000 0000. 0000 0000

2.0.0.0/24

pentru retea de 200 hosturi

2.0.1.0/24

$50+2 \leq 2^6 \Rightarrow$ avem nevoie de 6 biti de host

2.0.1. 0000 0000

01

10

11

Hosturi

200

50

30

20

7

4

2

2

2.0.0.0/23

$200+2 \leq 2^8 \Rightarrow$ avem nevoie de 8 biti de host

2.0. 0000 0000. 0000 0000

2.0.0.0/24

pentru retea de 200 hosturi

2.0.1.0/24

$50+2 \leq 2^6 \Rightarrow$ avem nevoie de 6 biti de host

2.0.1. 0000 0000

2.0.1.0/26

pentru retea de 50 hosturi

01

2.0.1.64/26

10

2.0.1.128/26

11

2.0.1.192/26

Hosturi

200

50

30

20

7

4

2

2

2.0.0.0/23

$200+2 \leq 2^8 \Rightarrow$ avem nevoie de 8 biti de host

2.0. 0000 0000. 0000 0000

2.0.0.0/24

pentru retea de 200 hosturi

2.0.1.0/24

$50+2 \leq 2^6 \Rightarrow$ avem nevoie de 6 biti de host

2.0.1. 0000 0000

2.0.1.0/26

pentru retea de 50 hosturi

01

2.0.1.64/26

10

2.0.1.128/26

11

2.0.1.192/26

$30+2 \leq 2^5 \Rightarrow$ avem nevoie de 5 biti de host

Hosturi

200

50

30

20

7

4

2

2

2.0.0.0/23

$200+2 \leq 2^8 \Rightarrow$ avem nevoie de 8 biti de host

2.0. 0000 0000. 0000 0000

2.0.0.0/24

pentru retea de 200 hosturi

2.0.1.0/24

$50+2 \leq 2^6 \Rightarrow$ avem nevoie de 6 biti de host

2.0.1. 0000 0000

2.0.1.0/26

pentru retea de 50 hosturi

2.0.1.64/26

2.0.1.128/26

2.0.1.192/26

$30+2 \leq 2^5 \Rightarrow$ avem nevoie de 5 biti de host

2.0.1.0100 0000

Hosturi

200

50

30

20

7

4

2

2

2.0.0.0/23

$200+2 \leq 2^8 \Rightarrow$ avem nevoie de 8 biti de host

2.0. 0000 0000 | 0000 0000

2.0.0.0/24

pentru rețeaua de 200 hosturi

2.0.1.0/24

$50+2 \leq 2^6 \Rightarrow$ avem nevoie de 6 biti de host

2.0.1. 0000 0000

2.0.1.0/26

pentru rețeaua de 50 hosturi

2.0.1.64/26

2.0.1.128/26

2.0.1.192/26

$30+2 \leq 2^5 \Rightarrow$ avem nevoie de 5 biti de host

2.0.1.0100 0000

1

Hosturi

200

50

30

20

7

4

2

2

2.0.0.0/23

$200+2 \leq 2^8 \Rightarrow$ avem nevoie de 8 biti de host

2.0. 0000 0000 | 0000 0000

2.0.0.0/24

pentru retea de 200 hosturi

2.0.1.0/24

$50+2 \leq 2^6 \Rightarrow$ avem nevoie de 6 biti de host

2.0.1. 0000 0000

2.0.1.0/26

pentru retea de 50 hosturi

2.0.1.64/26

2.0.1.128/26

2.0.1.192/26

$30+2 \leq 2^5 \Rightarrow$ avem nevoie de 5 biti de host

2.0.1.0100 0000

2.0.1.64/27

pentru retea de 30 hosturi

2.0.1.96/27

Hosturi

200

50

30

20

7

4

2

2

2.0.0.0/23

$200+2 \leq 2^8 \Rightarrow$ avem nevoie de 8 biti de host

2.0. 0000 0000 | 0000 0000

2.0.0.0/24

pentru rețeaua de 200 hosturi

2.0.1.0/24

$50+2 \leq 2^6 \Rightarrow$ avem nevoie de 6 biti de host

2.0.1. 0000 0000

2.0.1.0/26

pentru rețeaua de 50 hosturi

2.0.1.64/26

2.0.1.128/26

2.0.1.192/26

$30+2 \leq 2^5 \Rightarrow$ avem nevoie de 5 biti de host

2.0.1.0100 0000

2.0.1.64/27

pentru rețeaua de 30 hosturi

2.0.1.96/27

$20+2 \leq 2^5 \Rightarrow$ avem nevoie de 5 biti de host

Hosturi

200

50

30

20

7

4

2

2

2.0.0.0/23

$200+2 \leq 2^8 \Rightarrow$ avem nevoie de 8 biti de host

2.0. 0000 0000. 0000 0000

2.0.0.0/24

pentru retea de 200 hosturi

2.0.1.0/24

$50+2 \leq 2^6 \Rightarrow$ avem nevoie de 6 biti de host

2.0.1. 0000 0000

2.0.1.0/26

pentru retea de 50 hosturi

2.0.1.64/26

2.0.1.128/26

2.0.1.192/26

$30+2 \leq 2^5 \Rightarrow$ avem nevoie de 5 biti de host

2.0.1.0100 0000

2.0.1.64/27

pentru retea de 30 hosturi

2.0.1.96/27

$20+2 \leq 2^5 \Rightarrow$ avem nevoie de 5 biti de host

!!! Avem deja o retea /27 – cu 5 biti de host !!!

Hosturi

200

50

30

20

7

4

2

2

2.0.0.0/23

$200+2 \leq 2^8 \Rightarrow$ avem nevoie de 8 biti de host

2.0. 0000 0000 | 0000 0000

2.0.0.0/24

pentru rețeaua de 200 hosturi

2.0.1.0/24

$50+2 \leq 2^6 \Rightarrow$ avem nevoie de 6 biti de host

2.0.1. 0000 0000

2.0.1.0/26

pentru rețeaua de 50 hosturi

2.0.1.64/26

2.0.1.128/26

2.0.1.192/26

$30+2 \leq 2^5 \Rightarrow$ avem nevoie de 5 biti de host

2.0.1.0100 0000

2.0.1.64/27

pentru rețeaua de 30 hosturi

2.0.1.96/27

pentru rețeaua de 20 hosturi

$20+2 \leq 2^5 \Rightarrow$ avem nevoie de 5 biti de host

!!! Avem deja o rețea /27 – cu 5 biti de host !!!

Hosturi

200
50
30
20
7
4
2
2

2.0.0.0/23

$200+2 \leq 2^8 \Rightarrow$ avem nevoie de 8 biti de host

2.0. 0000 0000 | 0000 0000

2.0.0.0/24

pentru retea de 200 hosturi

2.0.1.0/24

$50+2 \leq 2^6 \Rightarrow$ avem nevoie de 6 biti de host

2.0.1. 0000 0000

2.0.1.0/26

pentru retea de 50 hosturi

2.0.1.64/26

01

2.0.1.128/26

10

2.0.1.192/26

11

$30+2 \leq 2^5 \Rightarrow$ avem nevoie de 5 biti de host

2.0.1.0100 0000

2.0.1.64/27

pentru retea de 30 hosturi

2.0.1.96/27

pentru retea de 20 hosturi

$20+2 \leq 2^5 \Rightarrow$ avem nevoie de 5 biti de host

!!! Avem deja o retea /27 – cu 5 biti de host !!!

$7+2 \leq 2^4 \Rightarrow$ avem nevoie de 4 biti de host

Hosturi

200

50

30

20

7

4

2

2

2.0.0.0/23

$200+2 \leq 2^8 \Rightarrow$ avem nevoie de 8 biti de host

2.0. 0000 0000. 0000 0000

2.0.0.0/24

pentru retea de 200 hosturi

2.0.1.0/24

$50+2 \leq 2^6 \Rightarrow$ avem nevoie de 6 biti de host

2.0.1. 0000 0000

2.0.1.0/26

pentru retea de 50 hosturi

01

2.0.1.64/26

10

2.0.1.128/26

11

2.0.1.192/26

$30+2 \leq 2^5 \Rightarrow$ avem nevoie de 5 biti de host

2.0.1.0100 0000

2.0.1.64/27

pentru retea de 30 hosturi

1

2.0.1.96/27

pentru retea de 20 hosturi

$20+2 \leq 2^5 \Rightarrow$ avem nevoie de 5 biti de host

!!! Avem deja o retea /27 – cu 5 biti de host !!!

$7+2 \leq 2^4 \Rightarrow$ avem nevoie de 4 biti de host

2.0.1.1000 0000

Hosturi

200

50

30

20

7

4

2

2

2.0.0.0/23

$200+2 \leq 2^8 \Rightarrow$ avem nevoie de 8 biti de host

2.0. 0000 0000 | 0000 0000

2.0.0.0/24

pentru retea de 200 hosturi

2.0.1.0/24

$50+2 \leq 2^6 \Rightarrow$ avem nevoie de 6 biti de host

2.0.1. 0000 0000

2.0.1.0/26

pentru retea de 50 hosturi

2.0.1.64/26

2.0.1.128/26

2.0.1.192/26

$30+2 \leq 2^5 \Rightarrow$ avem nevoie de 5 biti de host

2.0.1.0100 0000

2.0.1.64/27

pentru retea de 30 hosturi

2.0.1.96/27

pentru retea de 20 hosturi

$20+2 \leq 2^5 \Rightarrow$ avem nevoie de 5 biti de host

!!! Avem deja o retea /27 – cu 5 biti de host !!!

$7+2 \leq 2^4 \Rightarrow$ avem nevoie de 4 biti de host

2.0.1.1000 0000

01

10

11

Hosturi

200

50

30

20

7

4

2

2

2.0.0.0/23

$200+2 \leq 2^8 \Rightarrow$ avem nevoie de 8 biti de host

2.0. 0000 0000 | 0000 0000

2.0.0.0/24

pentru retea de 200 hosturi

2.0.1.0/24

$50+2 \leq 2^6 \Rightarrow$ avem nevoie de 6 biti de host

2.0.1. 0000 0000

2.0.1.0/26

pentru retea de 50 hosturi

01

2.0.1.64/26

10

2.0.1.128/26

11

2.0.1.192/26

$30+2 \leq 2^5 \Rightarrow$ avem nevoie de 5 biti de host

2.0.1.0100 0000

2.0.1.64/27

pentru retea de 30 hosturi

1

2.0.1.96/27

pentru retea de 20 hosturi

$20+2 \leq 2^5 \Rightarrow$ avem nevoie de 5 biti de host

!!! Avem deja o retea /27 – cu 5 biti de host !!!

$7+2 \leq 2^4 \Rightarrow$ avem nevoie de 4 biti de host

2.0.1.1000 0000

2.0.1.128/28

pentru retea de 7 hosturi

01

2.0.1.144/28

10

2.0.1.160/28

11

2.0.1.176/28

Hosturi

200

50

30

20

7

4

2

2

2.0.0.0/23

$200+2 \leq 2^8 \Rightarrow$ avem nevoie de 8 biti de host

2.0. 0000 0000 | 0000 0000

2.0.0.0/24

pentru retea de 200 hosturi

2.0.1.0/24

$50+2 \leq 2^6 \Rightarrow$ avem nevoie de 6 biti de host

2.0.1. 0000 0000

2.0.1.0/26

pentru retea de 50 hosturi

01

2.0.1.64/26

10

2.0.1.128/26

11

2.0.1.192/26

$30+2 \leq 2^5 \Rightarrow$ avem nevoie de 5 biti de host

2.0.1.0100 0000

2.0.1.64/27

pentru retea de 30 hosturi

1

2.0.1.96/27

pentru retea de 20 hosturi

$20+2 \leq 2^5 \Rightarrow$ avem nevoie de 5 biti de host

!!! Avem deja o retea /27 – cu 5 biti de host !!!

$7+2 \leq 2^4 \Rightarrow$ avem nevoie de 4 biti de host

2.0.1.1000 0000

2.0.1.128/28

pentru retea de 7 hosturi

01

2.0.1.144/28

10

2.0.1.160/28

11

2.0.1.176/28

$4+2 \leq 2^3 \Rightarrow$ avem nevoie de 3 biti de host

Hosturi

200
50
30
20
7
4
2
2

2.0.0.0/23

$200+2 \leq 2^8 \Rightarrow$ avem nevoie de 8 biti de host

2.0. 0000 0000 | 0000 0000

2.0.0.0/24

pentru retea de 200 hosturi

2.0.1.0/24

$50+2 \leq 2^6 \Rightarrow$ avem nevoie de 6 biti de host

2.0.1. 0000 0000

2.0.1.0/26

pentru retea de 50 hosturi

01

2.0.1.64/26

10

2.0.1.128/26

11

2.0.1.192/26

$30+2 \leq 2^5 \Rightarrow$ avem nevoie de 5 biti de host

2.0.1.0100 0000

2.0.1.64/27

pentru retea de 30 hosturi

1

2.0.1.96/27

pentru retea de 20 hosturi

$20+2 \leq 2^5 \Rightarrow$ avem nevoie de 5 biti de host

!!! Avem deja o retea /27 – cu 5 biti de host !!!

$7+2 \leq 2^4 \Rightarrow$ avem nevoie de 4 biti de host

2.0.1.1000 0000

2.0.1.128/28

pentru retea de 7 hosturi

01

2.0.1.144/28

10

2.0.1.160/28

11

2.0.1.176/28

$4+2 \leq 2^3 \Rightarrow$ avem nevoie de 3 biti de host

2.0.1.1001 0000

Hosturi

200

50

30

20

7

4

2

2

2.0.0.0/23

$200+2 \leq 2^8 \Rightarrow$ avem nevoie de 8 biti de host

2.0. 0000 0000 | 0000 0000

2.0.0.0/24

pentru retea de 200 hosturi

2.0.1.0/24

$50+2 \leq 2^6 \Rightarrow$ avem nevoie de 6 biti de host

2.0.1. 0000 0000

2.0.1.0/26

pentru retea de 50 hosturi

01

2.0.1.64/26

10

2.0.1.128/26

11

2.0.1.192/26

$30+2 \leq 2^5 \Rightarrow$ avem nevoie de 5 biti de host

2.0.1.0100 0000

2.0.1.64/27

pentru retea de 30 hosturi

1

2.0.1.96/27

pentru retea de 20 hosturi

$20+2 \leq 2^5 \Rightarrow$ avem nevoie de 5 biti de host

!!! Avem deja o retea /27 – cu 5 biti de host !!!

$7+2 \leq 2^4 \Rightarrow$ avem nevoie de 4 biti de host

2.0.1.1000 0000

2.0.1.128/28

pentru retea de 7 hosturi

01

2.0.1.144/28

10

2.0.1.160/28

11

2.0.1.176/28

$4+2 \leq 2^3 \Rightarrow$ avem nevoie de 3 biti de host

2.0.1.1001 0000

1

Hosturi

200

50

30

20

7

4

2

2

2.0.0.0/23

$200+2 \leq 2^8 \Rightarrow$ avem nevoie de 8 biti de host

2.0. 0000 0000 | 0000 0000

2.0.0.0/24

pentru retea de 200 hosturi

2.0.1.0/24

$50+2 \leq 2^6 \Rightarrow$ avem nevoie de 6 biti de host

2.0.1. 0000 0000

2.0.1.0/26

pentru retea de 50 hosturi

01

2.0.1.64/26

10

2.0.1.128/26

11

2.0.1.192/26

$30+2 \leq 2^5 \Rightarrow$ avem nevoie de 5 biti de host

2.0.1.0100 0000

2.0.1.64/27

pentru retea de 30 hosturi

1

2.0.1.96/27

pentru retea de 20 hosturi

$20+2 \leq 2^5 \Rightarrow$ avem nevoie de 5 biti de host

!!! Avem deja o retea /27 – cu 5 biti de host !!!

$7+2 \leq 2^4 \Rightarrow$ avem nevoie de 4 biti de host

2.0.1.1000 0000

2.0.1.128/28

pentru retea de 7 hosturi

01

2.0.1.144/28

10

2.0.1.160/28

11

2.0.1.176/28

$4+2 \leq 2^3 \Rightarrow$ avem nevoie de 3 biti de host

2.0.1.1001 0000

2.0.1.144/29

pentru retea de 4 hosturi

1

2.0.1.152/29

Hosturi

200

50

30

20

7

4

2

2

2.0.0.0/23

$200+2 \leq 2^8 \Rightarrow$ avem nevoie de 8 biti de host

2.0. 0000 0000 | 0000 0000

2.0.0.0/24

pentru retea de 200 hosturi

2.0.1.0/24

$50+2 \leq 2^6 \Rightarrow$ avem nevoie de 6 biti de host

2.0.1. 0000 0000

2.0.1.0/26

pentru retea de 50 hosturi

2.0.1.64/26

2.0.1.128/26

2.0.1.192/26

$30+2 \leq 2^5 \Rightarrow$ avem nevoie de 5 biti de host

2.0.1.0100 0000

2.0.1.64/27

pentru retea de 30 hosturi

2.0.1.96/27

pentru retea de 20 hosturi

$20+2 \leq 2^5 \Rightarrow$ avem nevoie de 5 biti de host

!!! Avem deja o retea /27 – cu 5 biti de host !!!

$7+2 \leq 2^4 \Rightarrow$ avem nevoie de 4 biti de host

2.0.1.1000 0000

2.0.1.128/28

pentru retea de 7 hosturi

2.0.1.144/28

2.0.1.160/28

2.0.1.176/28

$4+2 \leq 2^3 \Rightarrow$ avem nevoie de 3 biti de host

2.0.1.1001 0000

2.0.1.144/29

pentru retea de 4 hosturi

2.0.1.152/29

$2+2 \leq 2^2 \Rightarrow$ avem nevoie de 2 biti de host

Hosturi

200

50

30

20

7

4

2

2

2.0.0.0/23

$200+2 \leq 2^8 \Rightarrow$ avem nevoie de 8 biti de host

2.0. 0000 0000 | 0000 0000

2.0.0.0/24

pentru retea de 200 hosturi

2.0.1.0/24

$50+2 \leq 2^6 \Rightarrow$ avem nevoie de 6 biti de host

2.0.1. 0000 0000

2.0.1.0/26

pentru retea de 50 hosturi

2.0.1.64/26

2.0.1.128/26

2.0.1.192/26

$30+2 \leq 2^5 \Rightarrow$ avem nevoie de 5 biti de host

2.0.1.0100 0000

2.0.1.64/27

pentru retea de 30 hosturi

2.0.1.96/27

pentru retea de 20 hosturi

$20+2 \leq 2^5 \Rightarrow$ avem nevoie de 5 biti de host

!!! Avem deja o retea /27 – cu 5 biti de host !!!

$7+2 \leq 2^4 \Rightarrow$ avem nevoie de 4 biti de host

2.0.1.1000 0000

2.0.1.128/28

pentru retea de 7 hosturi

2.0.1.144/28

2.0.1.160/28

2.0.1.176/28

$4+2 \leq 2^3 \Rightarrow$ avem nevoie de 3 biti de host

2.0.1.1001 0000

2.0.1.144/29

pentru retea de 4 hosturi

2.0.1.152/29

$2+2 \leq 2^2 \Rightarrow$ avem nevoie de 2 biti de host

2.0.1.1001 1000

Hosturi

200
50
30
20
7
4
2
2

2.0.0.0/23

$200+2 \leq 2^8 \Rightarrow$ avem nevoie de 8 biti de host

2.0. 0000 0000 | 0000 0000

2.0.0.0/24

pentru retea de 200 hosturi

2.0.1.0/24

$50+2 \leq 2^6 \Rightarrow$ avem nevoie de 6 biti de host

2.0.1. 0000 0000

2.0.1.0/26

pentru retea de 50 hosturi

01

2.0.1.64/26

10

2.0.1.128/26

11

2.0.1.192/26

$30+2 \leq 2^5 \Rightarrow$ avem nevoie de 5 biti de host

2.0.1.0100 0000

2.0.1.64/27

pentru retea de 30 hosturi

1

2.0.1.96/27

pentru retea de 20 hosturi

$20+2 \leq 2^5 \Rightarrow$ avem nevoie de 5 biti de host

!!! Avem deja o retea /27 – cu 5 biti de host !!!

$7+2 \leq 2^4 \Rightarrow$ avem nevoie de 4 biti de host

2.0.1.1000 0000

2.0.1.128/28

pentru retea de 7 hosturi

01

2.0.1.144/28

10

2.0.1.160/28

11

2.0.1.176/28

$4+2 \leq 2^3 \Rightarrow$ avem nevoie de 3 biti de host

2.0.1.1001 0000

2.0.1.144/29

pentru retea de 4 hosturi

1

2.0.1.152/29

$2+2 \leq 2^2 \Rightarrow$ avem nevoie de 2 biti de host

2.0.1.1001 1000

1

Hosturi

200

50

30

20

7

4

2

2

2.0.0.0/23

$200+2 \leq 2^8 \Rightarrow$ avem nevoie de 8 biti de host

2.0. 0000 0000 | 0000 0000

2.0.0.0/24

pentru retea de 200 hosturi

2.0.1.0/24

$50+2 \leq 2^6 \Rightarrow$ avem nevoie de 6 biti de host

2.0.1. 0000 0000

2.0.1.0/26

pentru retea de 50 hosturi

01

2.0.1.64/26

10

2.0.1.128/26

11

2.0.1.192/26

$30+2 \leq 2^5 \Rightarrow$ avem nevoie de 5 biti de host

2.0.1.0100 0000

2.0.1.64/27

pentru retea de 30 hosturi

1

2.0.1.96/27

pentru retea de 20 hosturi

$20+2 \leq 2^5 \Rightarrow$ avem nevoie de 5 biti de host

!!! Avem deja o retea /27 – cu 5 biti de host !!!

$7+2 \leq 2^4 \Rightarrow$ avem nevoie de 4 biti de host

2.0.1.1000 0000

2.0.1.128/28

pentru retea de 7 hosturi

01

2.0.1.144/28

10

2.0.1.160/28

11

2.0.1.176/28

$4+2 \leq 2^3 \Rightarrow$ avem nevoie de 3 biti de host

2.0.1.1001 0000

2.0.1.144/29

pentru retea de 4 hosturi

1

2.0.1.152/29

$2+2 \leq 2^2 \Rightarrow$ avem nevoie de 2 biti de host

2.0.1.1001 1000

2.0.1.152/30

pentru prima retea de 2 hosturi

1

2.0.1.156/30

pentru a doua retea de 2 hosturi

Hosturi

200

50

30

20

7

4

2

2

Reteaua de 30 Hosturi: [2.0.1.64/27](#)

2.0.1. 0100 0000



Hosturi

200

50

30

20

7

4

2

2

Reteaua de 30 Hosturi: [2.0.1.64/27](#)

2.0.1. 0100 0000 2.0.1.64

0 0001 2.0.1.65

0 0010 2.0.1.66

...

1 1101 2.0.1.

1 1110

1 1111

Hosturi

200

50

30

20

7

4

2

2

Reteaua de 30 Hosturi: [2.0.1.64/27](#)

2.0.1. 0100 0000	2.0.1.64
0 0001	2.0.1.65
0 0010	2.0.1.66
...	
1 1101	2.0.1.93
1 1110	2.0.1.94
1 1111	2.0.1.95

Hosturi

200

50

30

20

7

4

2

2

Reteaua de 30 Hosturi: [2.0.1.64/27](#)

2.0.1. 0100 0000

0 0001

0 0010

...

1 1101

1 1110

1 1111

2.0.1.64 - IP retea

2.0.1.65 - primul IP Alocabil (de obicei se foloseste ptr DG)

2.0.1.66

2.0.1.93 **5biti >> $2^5=32$ IP-uri >> $2^5-2=30$ Alocabile**

2.0.1.94

2.0.1.95 - IP Directed Broadcast

Hosturi

200

50

30

20

7

4

2

2

Reteaua de 30 Hosturi: [2.0.1.64/27](#)

2.0.1. 0100 0000

0 0001

0 0010

...

1 1101

1 1110

1 1111

2.0.1.64 - IP retea

2.0.1.65 - primul IP Alocabil (de obicei se foloseste ptr DG)

2.0.1.66

2.0.1.93 **5biti >> $2^5=32$ IP-uri >> $2^5-2=30$ Alocabile**

2.0.1.94

2.0.1.95 - IP Directed Broadcast

Hosturi

200

50

30

20

7

4

2

2

Reteaua de 30 Hosturi: [2.0.1.64/27](#)

2.0.1. 0100 0000

2.0.1.64 - IP retea

0 0001

2.0.1.65 - primul IP Alocabil (de obicei se foloseste ptr DG)

0 0010

2.0.1.66

...

1 1101

2.0.1.93 5biti >> $2^5=32$ IP-uri >> $2^5-2=30$ Alocabile

1 1110

2.0.1.94

1 1111

2.0.1.95 - IP Directed Broadcast

SM: /27= 1111 1111 . 1111 1111 . 1111 1111 . 1110 0000

= 255 . 255 . 255 . 224

Exercitiul 3

Subnetizati 3.0.0.0/16

Hosturi
20000
5000
300
20

$20000 + 2 \leq 2^{15} \Rightarrow$ avem nevoie de 15 biti de host

Hosturi

20000

5000

300

20

3.0.0.0/16

$20000 + 2 = 2^{15} \Rightarrow$ avem nevoie de 15 biti de host

3.0. 0000 0000 . 0

Hosturi

20000

5000

300

20

3.0.0.0/16

$20000 + 2 = 2^{15} \Rightarrow$ avem nevoie de 15 biti de host

3.0.0.0000 0000 . 0 3.0.0.0/17 pentru rețeaua de 20000 hosturi
11 3.0.128.0/17

Hosturi
20000
5000
300
20

3.0.0.0/16

$20000 + 2 = 2^{15} \Rightarrow$ avem nevoie de 15 biti de host

3.0.0000 0000 . 0 3.0.0.0/17 pentru rețeaua de 20000 hosturi
11 3.0.128.0/17

$5000 + 2 = 2^{13} \Rightarrow$ avem nevoie de 13 biti de host

Hosturi
20000
5000
300
20

3.0.0.0/16

$20000 + 2 = 2^{15} \Rightarrow$ avem nevoie de 15 biti de host

3.0.0000 0000 . 0 3.0.0.0/17 pentru rețeaua de 20000 hosturi

3.0.128.0/17

$5000 + 2 = 2^{13} \Rightarrow$ avem nevoie de 13 biti de host

3.0.1000 0000 . 0

Hosturi
20000
5000
300
20

3.0.0.0/16

$20000 + 2 = 2^{15} \Rightarrow$ avem nevoie de 15 biti de host

3.0.0000 0000 . 0 3.0.0.0/17 pentru rețeaua de 20000 hosturi
11

3.0.128.0/17

$5000 + 2 = 2^{13} \Rightarrow$ avem nevoie de 13 biti de host

3.0.1000 0000 . 0
01
10
11

Hosturi
20000
5000
300
20

3.0.0.0/16

$20000 + 2 = 2^{15}$ => avem nevoie de 15 biti de host

3.0.0000 0000 . 0 3.0.0.0/17 pentru rețeaua de 20000 hosturi
11
3.0.128.0/17

$5000 + 2 = 2^{13}$ => avem nevoie de 13 biti de host

3.0.1000 0000 . 0 3.0.128.0/19 pentru rețeaua de 5000 hosturi
01 3.0.160.0/19
10 3.0.192.0/19
11 3.0.224.0/19

Hosturi
20000
5000
300
20

3.0.0.0/16

$20000+2 \leq 2^{15} \Rightarrow$ avem nevoie de 15 biti de host

3.0.0000 0000 . 0 3.0.0.0/17 pentru rețeaua de 20000 hosturi
11

3.0.128.0/17

$5000+2 \leq 2^{13} \Rightarrow$ avem nevoie de 13 biti de host

3.0.1000 0000 . 0 3.0.128.0/19 pentru rețeaua de 5000 hosturi
01 3.0.160.0/19
10 3.0.192.0/19
11 3.0.224.0/19

$300+2 \leq 2^9 \Rightarrow$ avem nevoie de 9 biti de host

Hosturi
20000
5000
300
20

3.0.0.0/16

$20000+2 \leq 2^{15} \Rightarrow$ avem nevoie de 15 biti de host

3.0.0000 0000 . 0 3.0.0.0/17 pentru rețeaua de 20000 hosturi
11 3.0.128.0/17

$5000+2 \leq 2^{13} \Rightarrow$ avem nevoie de 13 biti de host

3.0.1000 0000 . 0 3.0.128.0/19 pentru rețeaua de 5000 hosturi
01 3.0.160.0/19
10 3.0.192.0/19
11 3.0.224.0/19

$300+2 \leq 2^9 \Rightarrow$ avem nevoie de 9 biti de host

3.0.1010 0000 . 0

Hosturi
20000
5000
300
20

3.0.0.0/16

$20000 + 2 = 2^{15} \Rightarrow$ avem nevoie de 15 biti de host

3.0.0000 0000 . 0 3.0.0.0/17 pentru rețeaua de 20000 hosturi
 3.0.128.0/17

$$5000 + 2 = 2^{13} \Rightarrow \text{avem nevoie de 13 biti de host}$$

3.0.1000 0000 . 0	3.0.128.0/19	pentru rețeaua de 5000 hosturi
01	3.0.160.0/19	
10	3.0.192.0/19	
11	3.0.224.0/19	

$300+2 \leq 2^9 \Rightarrow$ avem nevoie de 9 biti de host

3.0	1010 0000	0.0
	0 001	
	0 010	
	0 011	
	...	
	1 110	
	1 111	

Hosturi
20000
5000
300
20

$20000+2 < 2^{15} \Rightarrow$ avem nevoie de 15 biti de host

3.0. 0000 0000 . 0

3.0.0.0/17

pentru rețeaua de 20000 hosturi

3.0.128.0/17

3.0. 1000 0000 . 0

3.0.128.0/19

pentru rețeaua de 5000 hosturi

3.0.160.0/19

3.0.192.0/19

3.0.224.0/19

3.0. 1010 0000 . 0

3.0.160.0/23

pentru rețeaua de 300 hosturi

3.0.162.0/23

3.0.164.0/23

3.0.166.0/23

■ ■ ■

3.0.188.0/23

3.0.190.0/23

0 00 1

0 010

0 01 1

...

$$1 \ 110$$

1 1 1 1

$20000+2 < 2^{15} \Rightarrow$ avem nevoie de 15 biti de host

3.0. 0000 0000 . 0

3.0.0.0/17

pentru rețeaua de 20000 hosturi

3.0.128.0/17

3.0. 1000 0000 . 0

3.0.128.0/19

pentru rețeaua de 5000 hosturi

3.0.160.0/19

3.0.192.0/19

3.0.224.0/19

3.0. 1010 0000 . 0

3.0.160.0/23

pentru rețeaua de 300 hosturi

3.0.162.0/23

3.0.164.0/23

3.0.166.0/23

■ ■ ■

3.0.188.0/23

3.0.190.0/23

$$20+2=<2^5$$

st

3.0. 0000 0000 . 0 3.0.0.0/17 pentru rețeaua de 20000 hosturi
1 3.0.128.0/17

3.0.1000 0000 . 0	3.0.128.0/19	pentru rețeaua de 5000 hosturi
01	3.0.160.0/19	
10	3.0.192.0/19	
11	3.0.224.0/19	

3.0. 1010 0000 . 0	3.0.160.0/23	pentru rețeaua de 300 hosturi
0 001	3.0.162.0/23	
0 010	3.0.164.0/23	
0 011	3.0.166.0/23	
...	...	
1 110	3.0.188.0/23	
1 111	3.0.190.0/23	

3.0. 1010 001 | 0 . 0000 0000



3.0.0000 0000 . 0 3.0.0.0/17 pentru rețeaua de 20000 hosturi
1 3.0.128.0/17

3.0.1000 0000 . 0	3.0.128.0/19	pentru rețeaua de 5000 hosturi
01	3.0.160.0/19	
10	3.0.192.0/19	
11	3.0.224.0/19	

3.0. 1010 0000 . 0	3.0.160.0/23	pentru rețeaua de 300 hosturi
0 001	3.0.162.0/23	
0 010	3.0.164.0/23	
0 011	3.0.166.0/23	
...	...	
1 110	3.0.188.0/23	
1 111	3.0.190.0/23	

3.0. 1010 001	0 . 000	0 0000
	0 . 001	
	0 . 010	
	0 . 011	
	...	
	1 . 110	
	1 . 111	

$20000+2 < 2^{15} \Rightarrow$ avem nevoie de 15 biti de host

3.0. 0000 0000 . 0

3.0.0.0/17

pentru rețeaua de 20000 hosturi

3.0.128.0/17

3.0. 1000 0000 . 0

3.0.128.0/19

pentru rețeaua de 5000 hosturi

3.0.160.0/19

3.0.192.0/19

3.0.224.0/19

3.0. 1010 000b . 0

3.0.160.0/23

pentru rețeaua de 300 hosturi

3.0.162.0/23

3.0.164.0/23

3.0.166.0/23

010110010, $\equiv 0$

3.0.188.0/23

3 0 190 0/23

3.0. 1010 001|0 . 000|0 0000

3.0.162.0/27

pentru rețeaua de 20

hosturi

0 001

30 162 32/27

0.010

3.0 162 64/27

0.010
0.011

3.0.162.06/27

0.011

3.0.162.96/27

1 ... 110

0 0 100 100/0

1. 110

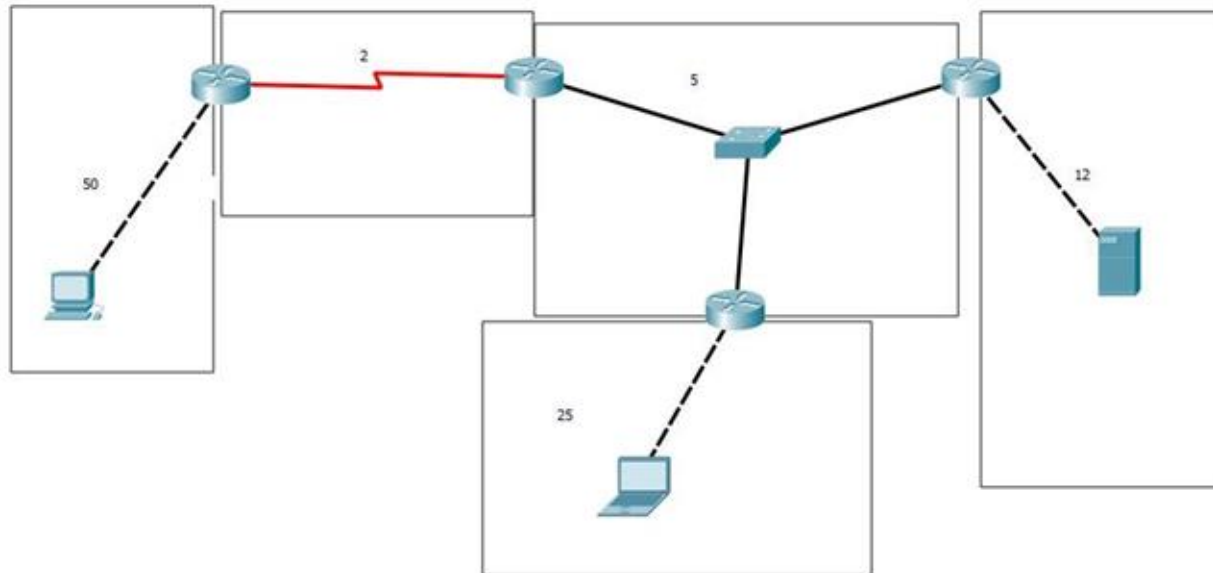
3.0.163.192/2

1. 111

3.0.163.224/2

Exercitiul 4

Subnetizati 4.0.0.0/24



Hosturi
50
25
12
5
2

4.0.0.0/25

4.0.0.0/24

4.0.0.128/25

Hosturi
50
25
12
5
2

4.0.0.0/26 pentru rețeaua de 50 hosturi

4.0.0.0/25

4.0.0.64/26

4.0.0.0/24

4.0.0.128/25

Hosturi
50
25
12
5
2

4.0.0.0/26 pentru rețeaua de 50 hosturi

4.0.0.0/25

4.0.0.64/27 pentru rețeaua de 25 hosturi

4.0.0.64/26

4.0.0.0/24

4.0.0.96/27

4.0.0.128/25

Hosturi
50
25
12
5
2

4.0.0.0/26 pentru rețeaua de 50 hosturi

4.0.0.0/25

4.0.0.64/27 pentru rețeaua de 25 hosturi

4.0.0.64/26

4.0.0.96/28 pentru rețeaua de 12 hosturi

4.0.0.0/24

4.0.0.96/27

4.0.0.112/28

4.0.0.128/25

Hosturi
50
25
12
5
2

4.0.0.0/26 pentru rețeaua de 50 hosturi

4.0.0.0/25 4.0.0.64/27 pentru rețeaua de 25 hosturi

4.0.0.64/26 4.0.0.96/28 pentru rețeaua de 12 hosturi

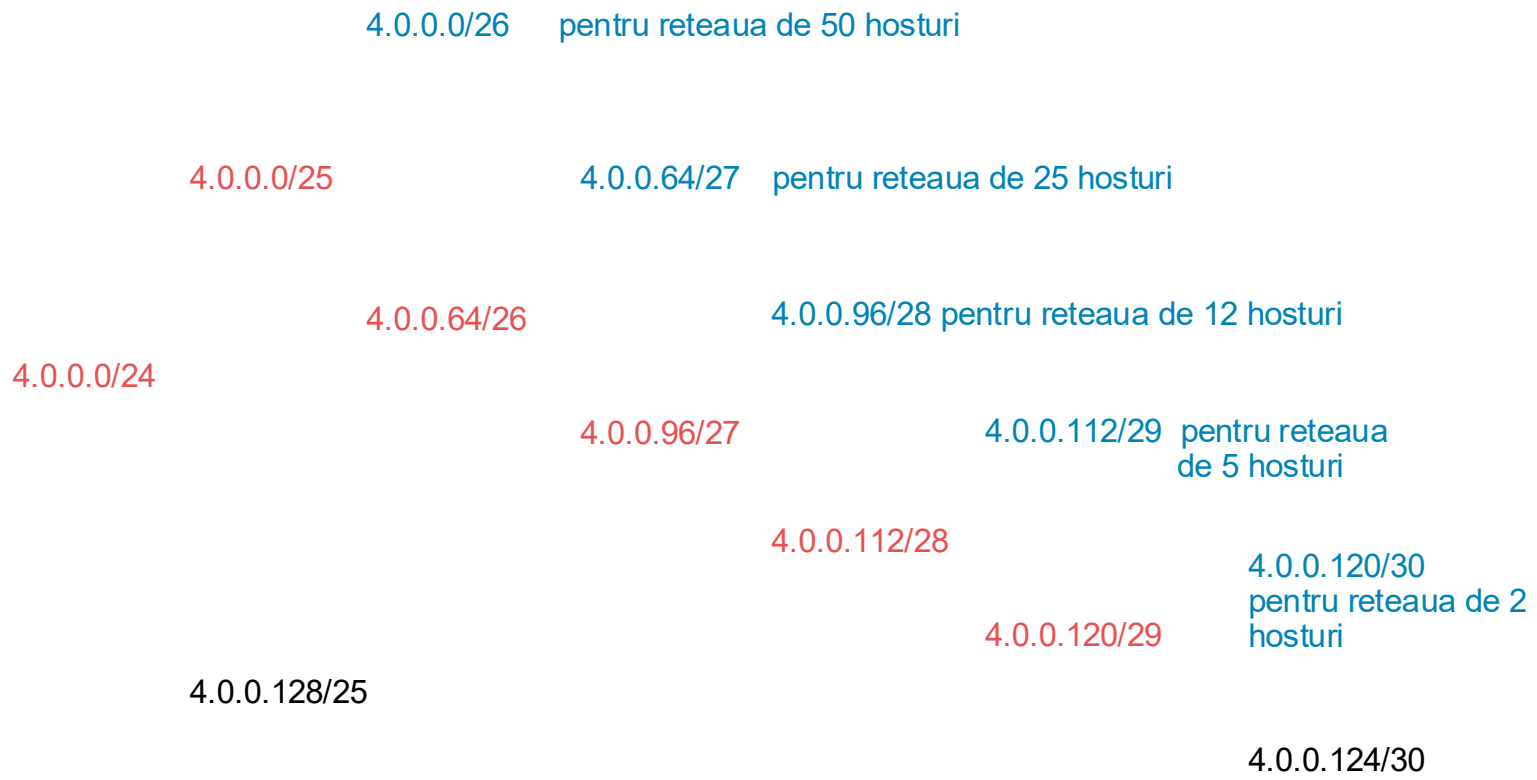
4.0.0.0/24 4.0.0.96/27 4.0.0.112/29 pentru rețeaua
de 5 hosturi

4.0.0.112/28

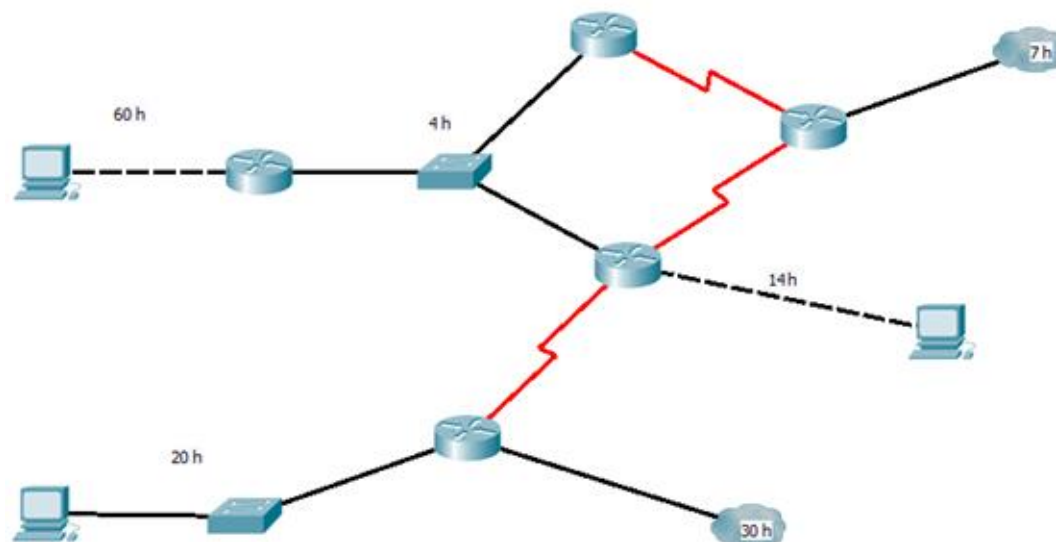
4.0.0.120/29

4.0.0.128/25

Hosturi
50
25
12
5
2



Exercitiul 5 de subnetizare:
Alocati adrese IP cat mai eficient folosind reteaua 5.0.0.0/24



5.0.0.0/24

Hosturi

$60 + 2$ (IP Retea Si IP DB) $\Rightarrow 2^6$ - Avem nevoie de 6 biti de host

60

31

20

14

8

4

2

2

2

Retea: 5.0.0.160/28

5.0.0.1010	0000	5.0.0.160
	0001	5.0.0.161
	0010	5.0.0.162
	0011	5.0.0.163
	...	
	1100	5.0.0.172
	1101	5.0.0.173
	1110	5.0.0.174
	1111	5.0.0.175

IP Retea
Primul Ip din retea .1 – IP Def Gateway

2^{4-2} IPuri alocabile

IP Directed Broadcast

	6.0.0.0/24	7.0.0.0/24	8.0.0.0/20
100	6.0.0.0	50	2000
30		40	1000
20		35	200
14		30	150
8		10	100
4		4	60
2		2	10
2		2	5

	6.0.0.0/24		7.0.0.0/24		8.0.0.0/20
100	6.0.0.0/25	50		2000	
30		40		1000	
20		35		200	
14		30		150	
8		10		100	
4		4		60	
2		2		10	
2		2		5	

	6.0.0.0/24		7.0.0.0/24		8.0.0.0/20
100	6.0.0.0/25	50		2000	
30	6.0.0.128	40		1000	
20		35		200	
14		30		150	
8		10		100	
4		4		60	
2		2		10	
2		2		5	

	6.0.0.0/24		7.0.0.0/24		8.0.0.0/20
100	6.0.0.0/25	50		2000	
30	6.0.0.128/27	40		1000	
20		35		200	
14		30		150	
8		10		100	
4		4		60	
2		2		10	
2		2		5	

	6.0.0.0/24		7.0.0.0/24		8.0.0.0/20
100	6.0.0.0/25	50		2000	
30	6.0.0.128/27	40		1000	
20	6.0.0.160	35		200	
14		30		150	
8		10		100	
4		4		60	
2		2		10	
2		2		5	

	6.0.0.0/24		7.0.0.0/24		8.0.0.0/20
100	6.0.0.0/25	50		2000	
30	6.0.0.128/27	40		1000	
20	6.0.0.160/27	35		200	
14		30		150	
8		10		100	
4		4		60	
2		2		10	
2		2		5	

	6.0.0.0/24		7.0.0.0/24		8.0.0.0/20
100	6.0.0.0/25	50		2000	
30	6.0.0.128/27	40		1000	
20	6.0.0.160/27	35		200	
14	6.0.0.192	30		150	
8		10		100	
4		4		60	
2		2		10	
2		2		5	

	6.0.0.0/24		7.0.0.0/24		8.0.0.0/20
100	6.0.0.0/25	50		2000	
30	6.0.0.128/27	40		1000	
20	6.0.0.160/27	35		200	
14	6.0.0.192/28	30		150	
8		10		100	
4		4		60	
2		2		10	
2		2		5	

	6.0.0.0/24		7.0.0.0/24		8.0.0.0/20
100	6.0.0.0/25	50		2000	
30	6.0.0.128/27	40		1000	
20	6.0.0.160/27	35		200	
14	6.0.0.192/28	30		150	
8	6.0.0.208	10		100	
4		4		60	
2		2		10	
2		2		5	

	6.0.0.0/24		7.0.0.0/24		8.0.0.0/20
100	6.0.0.0/25	50		2000	
30	6.0.0.128/27	40		1000	
20	6.0.0.160/27	35		200	
14	6.0.0.192/28	30		150	
8	6.0.0.208/28	10		100	
4		4		60	
2		2		10	
2		2		5	

	6.0.0.0/24		7.0.0.0/24		8.0.0.0/20
100	6.0.0.0/25	50		2000	
30	6.0.0.128/27	40		1000	
20	6.0.0.160/27	35		200	
14	6.0.0.192/28	30		150	
8	6.0.0.208/28	10		100	
4	6.0.0.224	4		60	
2		2		10	
2		2		5	

	6.0.0.0/24		7.0.0.0/24		8.0.0.0/20
100	6.0.0.0/25	50		2000	
30	6.0.0.128/27	40		1000	
20	6.0.0.160/27	35		200	
14	6.0.0.192/28	30		150	
8	6.0.0.208/28	10		100	
4	6.0.0.224/29	4		60	
2		2		10	
2		2		5	

	6.0.0.0/24	7.0.0.0/24	8.0.0.0/20
100	6.0.0.0/25	50	2000
30	6.0.0.128/27	40	1000
20	6.0.0.160/27	35	200
14	6.0.0.192/28	30	150
8	6.0.0.208/28	10	100
4	6.0.0.224/29	4	60
2	6.0.0.232	2	10
2		2	5

	6.0.0.0/24		7.0.0.0/24		8.0.0.0/20
100	6.0.0.0/25	50		2000	
30	6.0.0.128/27	40		1000	
20	6.0.0.160/27	35		200	
14	6.0.0.192/28	30		150	
8	6.0.0.208/28	10		100	
4	6.0.0.224/29	4		60	
2	6.0.0.232/30	2		10	
2		2		5	

	6.0.0.0/24	7.0.0.0/24	8.0.0.0/20
100	6.0.0.0/25	50	2000
30	6.0.0.128/27	40	1000
20	6.0.0.160/27	35	200
14	6.0.0.192/28	30	150
8	6.0.0.208/28	10	100
4	6.0.0.224/29	4	60
2	6.0.0.232/30	2	10
2	6.0.0.236	2	5

	6.0.0.0/24	7.0.0.0/24	8.0.0.0/20
100	6.0.0.0/25	50	2000
30	6.0.0.128/27	40	1000
20	6.0.0.160/27	35	200
14	6.0.0.192/28	30	150
8	6.0.0.208/28	10	100
4	6.0.0.224/29	4	60
2	6.0.0.232/30	2	10
2	6.0.0.236/30	2	5

	6.0.0.0/24		7.0.0.0/24		8.0.0.0/20
100	6.0.0.0/25	50	7.0.0.0	2000	
30	6.0.0.128/27	40		1000	
20	6.0.0.160/27	35		200	
14	6.0.0.192/28	30		150	
8	6.0.0.208/28	10		100	
4	6.0.0.224/29	4		60	
2	6.0.0.232/30	2		10	
2	6.0.0.236/30	2		5	

	6.0.0.0/24		7.0.0.0/24		8.0.0.0/20
100	6.0.0.0/25	50	7.0.0.0/26	2000	
30	6.0.0.128/27	40		1000	
20	6.0.0.160/27	35		200	
14	6.0.0.192/28	30		150	
8	6.0.0.208/28	10		100	
4	6.0.0.224/29	4		60	
2	6.0.0.232/30	2		10	
2	6.0.0.236/30	2		5	

	6.0.0.0/24		7.0.0.0/24	8.0.0.0/20
100	6.0.0.0/25	50	7.0.0.0/26	2000
30	6.0.0.128/27	40	7.0.0.64	1000
20	6.0.0.160/27	35		200
14	6.0.0.192/28	30		150
8	6.0.0.208/28	10		100
4	6.0.0.224/29	4		60
2	6.0.0.232/30	2		10
2	6.0.0.236/30	2		5

	6.0.0.0/24		7.0.0.0/24		8.0.0.0/20
100	6.0.0.0/25	50	7.0.0.0/26	2000	
30	6.0.0.128/27	40	7.0.0.64/26	1000	
20	6.0.0.160/27	35		200	
14	6.0.0.192/28	30		150	
8	6.0.0.208/28	10		100	
4	6.0.0.224/29	4		60	
2	6.0.0.232/30	2		10	
2	6.0.0.236/30	2		5	

	6.0.0.0/24		7.0.0.0/24	8.0.0.0/20
100	6.0.0.0/25	50	7.0.0.0/26	2000
30	6.0.0.128/27	40	7.0.0.64/26	1000
20	6.0.0.160/27	35	7.0.0.128	200
14	6.0.0.192/28	30		150
8	6.0.0.208/28	10		100
4	6.0.0.224/29	4		60
2	6.0.0.232/30	2		10
2	6.0.0.236/30	2		5

	6.0.0.0/24		7.0.0.0/24	8.0.0.0/20
100	6.0.0.0/25	50	7.0.0.0/26	2000
30	6.0.0.128/27	40	7.0.0.64/26	1000
20	6.0.0.160/27	35	7.0.0.128/26	200
14	6.0.0.192/28	30		150
8	6.0.0.208/28	10		100
4	6.0.0.224/29	4		60
2	6.0.0.232/30	2		10
2	6.0.0.236/30	2		5

	6.0.0.0/24		7.0.0.0/24		8.0.0.0/20
100	6.0.0.0/25	50	7.0.0.0/26	2000	
30	6.0.0.128/27	40	7.0.0.64/26	1000	
20	6.0.0.160/27	35	7.0.0.128/26	200	
14	6.0.0.192/28	30	7.0.0.192	150	
8	6.0.0.208/28	10		100	
4	6.0.0.224/29	4		60	
2	6.0.0.232/30	2		10	
2	6.0.0.236/30	2		5	

	6.0.0.0/24		7.0.0.0/24		8.0.0.0/20
100	6.0.0.0/25	50	7.0.0.0/26	2000	
30	6.0.0.128/27	40	7.0.0.64/26	1000	
20	6.0.0.160/27	35	7.0.0.128/26	200	
14	6.0.0.192/28	30	7.0.0.192/27	150	
8	6.0.0.208/28	10		100	
4	6.0.0.224/29	4		60	
2	6.0.0.232/30	2		10	
2	6.0.0.236/30	2		5	

	6.0.0.0/24		7.0.0.0/24		8.0.0.0/20
100	6.0.0.0/25	50	7.0.0.0/26	2000	
30	6.0.0.128/27	40	7.0.0.64/26	1000	
20	6.0.0.160/27	35	7.0.0.128/26	200	
14	6.0.0.192/28	30	7.0.0.192/27	150	
8	6.0.0.208/28	10	7.0.0.224	100	
4	6.0.0.224/29	4		60	
2	6.0.0.232/30	2		10	
2	6.0.0.236/30	2		5	

	6.0.0.0/24		7.0.0.0/24	8.0.0.0/20
100	6.0.0.0/25	50	7.0.0.0/26	2000
30	6.0.0.128/27	40	7.0.0.64/26	1000
20	6.0.0.160/27	35	7.0.0.128/26	200
14	6.0.0.192/28	30	7.0.0.192/27	150
8	6.0.0.208/28	10	7.0.0.224/28	100
4	6.0.0.224/29	4		60
2	6.0.0.232/30	2		10
2	6.0.0.236/30	2		5

	6.0.0.0/24		7.0.0.0/24	8.0.0.0/20
100	6.0.0.0/25	50	7.0.0.0/26	2000
30	6.0.0.128/27	40	7.0.0.64/26	1000
20	6.0.0.160/27	35	7.0.0.128/26	200
14	6.0.0.192/28	30	7.0.0.192/27	150
8	6.0.0.208/28	10	7.0.0.224/28	100
4	6.0.0.224/29	4	7.0.0.240	60
2	6.0.0.232/30	2		10
2	6.0.0.236/30	2		5

	6.0.0.0/24		7.0.0.0/24	8.0.0.0/20
100	6.0.0.0/25	50	7.0.0.0/26	2000
30	6.0.0.128/27	40	7.0.0.64/26	1000
20	6.0.0.160/27	35	7.0.0.128/26	200
14	6.0.0.192/28	30	7.0.0.192/27	150
8	6.0.0.208/28	10	7.0.0.224/28	100
4	6.0.0.224/29	4	7.0.0.240/29	60
2	6.0.0.232/30	2		10
2	6.0.0.236/30	2		5

	6.0.0.0/24		7.0.0.0/24	8.0.0.0/20
100	6.0.0.0/25	50	7.0.0.0/26	2000
30	6.0.0.128/27	40	7.0.0.64/26	1000
20	6.0.0.160/27	35	7.0.0.128/26	200
14	6.0.0.192/28	30	7.0.0.192/27	150
8	6.0.0.208/28	10	7.0.0.224/28	100
4	6.0.0.224/29	4	7.0.0.240/29	60
2	6.0.0.232/30	2	7.0.0.248	10
2	6.0.0.236/30	2		5

	6.0.0.0/24		7.0.0.0/24	8.0.0.0/20
100	6.0.0.0/25	50	7.0.0.0/26	2000
30	6.0.0.128/27	40	7.0.0.64/26	1000
20	6.0.0.160/27	35	7.0.0.128/26	200
14	6.0.0.192/28	30	7.0.0.192/27	150
8	6.0.0.208/28	10	7.0.0.224/28	100
4	6.0.0.224/29	4	7.0.0.240/29	60
2	6.0.0.232/30	2	7.0.0.248/30	10
2	6.0.0.236/30	2		5

	6.0.0.0/24		7.0.0.0/24	8.0.0.0/20
100	6.0.0.0/25	50	7.0.0.0/26	2000
30	6.0.0.128/27	40	7.0.0.64/26	1000
20	6.0.0.160/27	35	7.0.0.128/26	200
14	6.0.0.192/28	30	7.0.0.192/27	150
8	6.0.0.208/28	10	7.0.0.224/28	100
4	6.0.0.224/29	4	7.0.0.240/29	60
2	6.0.0.232/30	2	7.0.0.248/30	10
2	6.0.0.236/30	2	7.0.0.252	5

	6.0.0.0/24		7.0.0.0/24	8.0.0.0/20
100	6.0.0.0/25	50	7.0.0.0/26	2000
30	6.0.0.128/27	40	7.0.0.64/26	1000
20	6.0.0.160/27	35	7.0.0.128/26	200
14	6.0.0.192/28	30	7.0.0.192/27	150
8	6.0.0.208/28	10	7.0.0.224/28	100
4	6.0.0.224/29	4	7.0.0.240/29	60
2	6.0.0.232/30	2	7.0.0.248/30	10
2	6.0.0.236/30	2	7.0.0.252/30	5

	6.0.0.0/24		7.0.0.0/24		8.0.0.0/20
100	6.0.0.0/25	50	7.0.0.0/26	2000	8.0.0.0
30	6.0.0.128/27	40	7.0.0.64/26	1000	
20	6.0.0.160/27	35	7.0.0.128/26	200	
14	6.0.0.192/28	30	7.0.0.192/27	150	
8	6.0.0.208/28	10	7.0.0.224/28	100	
4	6.0.0.224/29	4	7.0.0.240/29	60	
2	6.0.0.232/30	2	7.0.0.248/30	10	
2	6.0.0.236/30	2	7.0.0.252	5	

	6.0.0.0/24		7.0.0.0/24		8.0.0.0/20
100	6.0.0.0/25	50	7.0.0.0/26	2000	8.0.0.0/21
30	6.0.0.128/27	40	7.0.0.64/26	1000	
20	6.0.0.160/27	35	7.0.0.128/26	200	
14	6.0.0.192/28	30	7.0.0.192/27	150	
8	6.0.0.208/28	10	7.0.0.224/28	100	
4	6.0.0.224/29	4	7.0.0.240/29	60	
2	6.0.0.232/30	2	7.0.0.248/30	10	
2	6.0.0.236/30	2	7.0.0.252	5	

	6.0.0.0/24		7.0.0.0/24		8.0.0.0/20
100	6.0.0.0/25	50	7.0.0.0/26	2000	8.0.0.0/21
30	6.0.0.128/27	40	7.0.0.64/26	1000	8.0.8.0
20	6.0.0.160/27	35	7.0.0.128/26	200	
14	6.0.0.192/28	30	7.0.0.192/27	150	
8	6.0.0.208/28	10	7.0.0.224/28	100	
4	6.0.0.224/29	4	7.0.0.240/29	60	
2	6.0.0.232/30	2	7.0.0.248/30	10	
2	6.0.0.236/30	2	7.0.0.252	5	

	6.0.0.0/24		7.0.0.0/24		8.0.0.0/20
100	6.0.0.0/25	50	7.0.0.0/26	2000	8.0.0.0/21
30	6.0.0.128/27	40	7.0.0.64/26	1000	8.0.8.0/22
20	6.0.0.160/27	35	7.0.0.128/26	200	
14	6.0.0.192/28	30	7.0.0.192/27	150	
8	6.0.0.208/28	10	7.0.0.224/28	100	
4	6.0.0.224/29	4	7.0.0.240/29	60	
2	6.0.0.232/30	2	7.0.0.248/30	10	
2	6.0.0.236/30	2	7.0.0.252	5	

	6.0.0.0/24		7.0.0.0/24		8.0.0.0/20
100	6.0.0.0/25	50	7.0.0.0/26	2000	8.0.0.0/21
30	6.0.0.128/27	40	7.0.0.64/26	1000	8.0.8.0/22
20	6.0.0.160/27	35	7.0.0.128/26	200	8.0.12.0
14	6.0.0.192/28	30	7.0.0.192/27	150	
8	6.0.0.208/28	10	7.0.0.224/28	100	
4	6.0.0.224/29	4	7.0.0.240/29	60	
2	6.0.0.232/30	2	7.0.0.248/30	10	
2	6.0.0.236/30	2	7.0.0.252	5	

	6.0.0.0/24		7.0.0.0/24		8.0.0.0/20
100	6.0.0.0/25	50	7.0.0.0/26	2000	8.0.0.0/21
30	6.0.0.128/27	40	7.0.0.64/26	1000	8.0.8.0/22
20	6.0.0.160/27	35	7.0.0.128/26	200	8.0.12.0/24
14	6.0.0.192/28	30	7.0.0.192/27	150	
8	6.0.0.208/28	10	7.0.0.224/28	100	
4	6.0.0.224/29	4	7.0.0.240/29	60	
2	6.0.0.232/30	2	7.0.0.248/30	10	
2	6.0.0.236/30	2	7.0.0.252	5	

	6.0.0.0/24		7.0.0.0/24		8.0.0.0/20
100	6.0.0.0/25	50	7.0.0.0/26	2000	8.0.0.0/21
30	6.0.0.128/27	40	7.0.0.64/26	1000	8.0.8.0/22
20	6.0.0.160/27	35	7.0.0.128/26	200	8.0.12.0/24
14	6.0.0.192/28	30	7.0.0.192/27	150	8.0.13.0
8	6.0.0.208/28	10	7.0.0.224/28	100	
4	6.0.0.224/29	4	7.0.0.240/29	60	
2	6.0.0.232/30	2	7.0.0.248/30	10	
2	6.0.0.236/30	2	7.0.0.252	5	

	6.0.0.0/24		7.0.0.0/24		8.0.0.0/20
100	6.0.0.0/25	50	7.0.0.0/26	2000	8.0.0.0/21
30	6.0.0.128/27	40	7.0.0.64/26	1000	8.0.8.0/22
20	6.0.0.160/27	35	7.0.0.128/26	200	8.0.12.0/24
14	6.0.0.192/28	30	7.0.0.192/27	150	8.0.13.0/24
8	6.0.0.208/28	10	7.0.0.224/28	100	
4	6.0.0.224/29	4	7.0.0.240/29	60	
2	6.0.0.232/30	2	7.0.0.248/30	10	
2	6.0.0.236/30	2	7.0.0.252	5	

	6.0.0.0/24		7.0.0.0/24		8.0.0.0/20
100	6.0.0.0/25	50	7.0.0.0/26	2000	8.0.0.0/21
30	6.0.0.128/27	40	7.0.0.64/26	1000	8.0.8.0/22
20	6.0.0.160/27	35	7.0.0.128/26	200	8.0.12.0/24
14	6.0.0.192/28	30	7.0.0.192/27	150	8.0.13.0/24
8	6.0.0.208/28	10	7.0.0.224/28	100	8.0.14.0
4	6.0.0.224/29	4	7.0.0.240/29	60	
2	6.0.0.232/30	2	7.0.0.248/30	10	
2	6.0.0.236/30	2	7.0.0.252	5	

	6.0.0.0/24		7.0.0.0/24		8.0.0.0/20
100	6.0.0.0/25	50	7.0.0.0/26	2000	8.0.0.0/21
30	6.0.0.128/27	40	7.0.0.64/26	1000	8.0.8.0/22
20	6.0.0.160/27	35	7.0.0.128/26	200	8.0.12.0/24
14	6.0.0.192/28	30	7.0.0.192/27	150	8.0.13.0/24
8	6.0.0.208/28	10	7.0.0.224/28	100	8.0.14.0/25
4	6.0.0.224/29	4	7.0.0.240/29	60	
2	6.0.0.232/30	2	7.0.0.248/30	10	
2	6.0.0.236/30	2	7.0.0.252	5	

	6.0.0.0/24		7.0.0.0/24		8.0.0.0/20
100	6.0.0.0/25	50	7.0.0.0/26	2000	8.0.0.0/21
30	6.0.0.128/27	40	7.0.0.64/26	1000	8.0.8.0/22
20	6.0.0.160/27	35	7.0.0.128/26	200	8.0.12.0/24
14	6.0.0.192/28	30	7.0.0.192/27	150	8.0.13.0/24
8	6.0.0.208/28	10	7.0.0.224/28	100	8.0.14.0/25
4	6.0.0.224/29	4	7.0.0.240/29	60	8.0.14.128
2	6.0.0.232/30	2	7.0.0.248/30	10	
2	6.0.0.236/30	2	7.0.0.252	5	

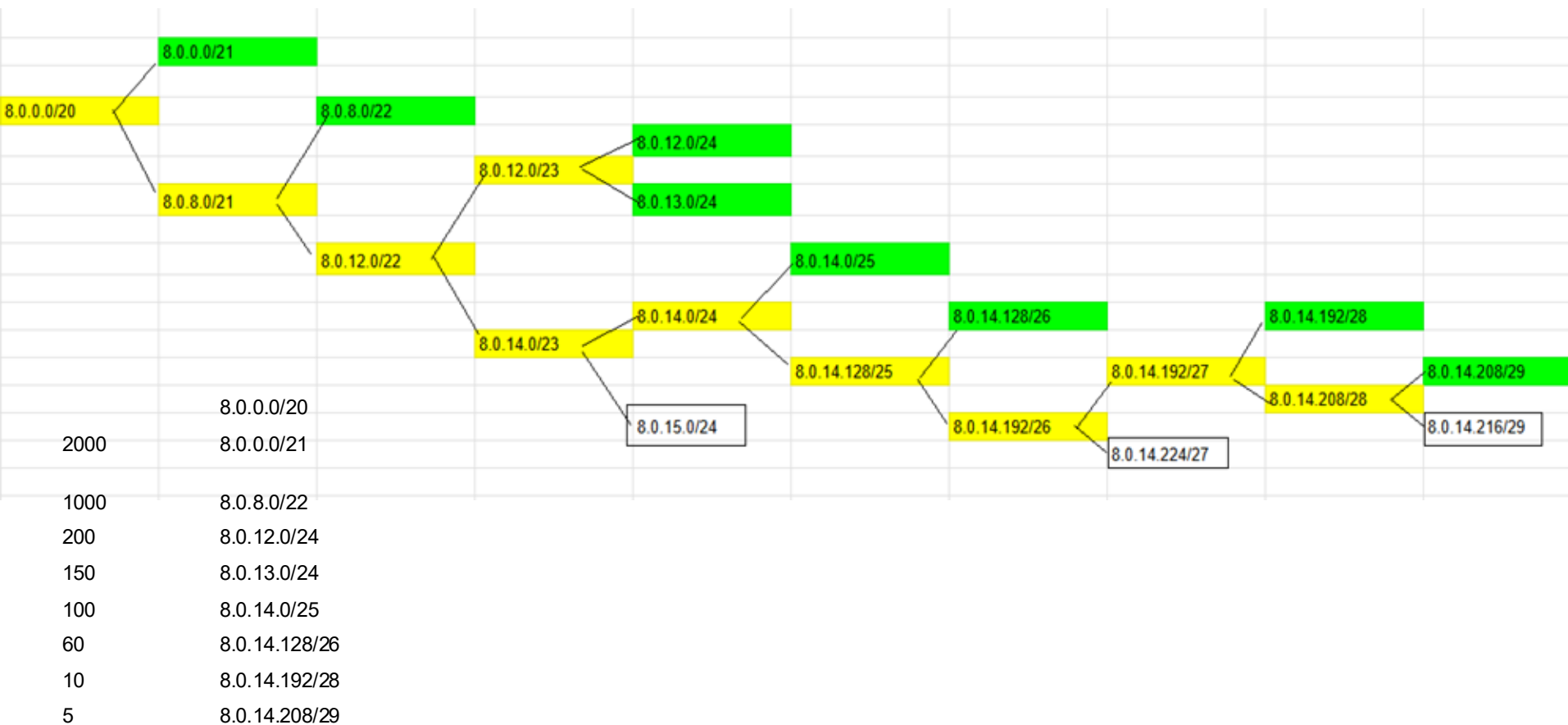
	6.0.0.0/24		7.0.0.0/24		8.0.0.0/20
100	6.0.0.0/25	50	7.0.0.0/26	2000	8.0.0.0/21
30	6.0.0.128/27	40	7.0.0.64/26	1000	8.0.8.0/22
20	6.0.0.160/27	35	7.0.0.128/26	200	8.0.12.0/24
14	6.0.0.192/28	30	7.0.0.192/27	150	8.0.13.0/24
8	6.0.0.208/28	10	7.0.0.224/28	100	8.0.14.0/25
4	6.0.0.224/29	4	7.0.0.240/29	60	8.0.14.128/2
2	6.0.0.232/30	2	7.0.0.248/30	6	
2	6.0.0.236/30	2	7.0.0.252	10	
				5	

	6.0.0.0/24		7.0.0.0/24		8.0.0.0/20
100	6.0.0.0/25	50	7.0.0.0/26	2000	8.0.0.0/21
30	6.0.0.128/27	40	7.0.0.64/26	1000	8.0.8.0/22
20	6.0.0.160/27	35	7.0.0.128/26	200	8.0.12.0/24
14	6.0.0.192/28	30	7.0.0.192/27	150	8.0.13.0/24
8	6.0.0.208/28	10	7.0.0.224/28	100	8.0.14.0/25
4	6.0.0.224/29	4	7.0.0.240/29	60	8.0.14.128/2
2	6.0.0.232/30	2	7.0.0.248/30	6	
2	6.0.0.236/30	2	7.0.0.252	10	8.0.14.192
				5	

	6.0.0.0/24		7.0.0.0/24		8.0.0.0/20
100	6.0.0.0/25	50	7.0.0.0/26	2000	8.0.0.0/21
30	6.0.0.128/27	40	7.0.0.64/26	1000	8.0.8.0/22
20	6.0.0.160/27	35	7.0.0.128/26	200	8.0.12.0/24
14	6.0.0.192/28	30	7.0.0.192/27	150	8.0.13.0/24
8	6.0.0.208/28	10	7.0.0.224/28	100	8.0.14.0/25
4	6.0.0.224/29	4	7.0.0.240/29	60	8.0.14.128/2
2	6.0.0.232/30	2	7.0.0.248/30	6	
2	6.0.0.236/30	2	7.0.0.252	10	8.0.14.192/2
				8	
				5	

	6.0.0.0/24		7.0.0.0/24		8.0.0.0/20
100	6.0.0.0/25	50	7.0.0.0/26	2000	8.0.0.0/21
30	6.0.0.128/27	40	7.0.0.64/26	1000	8.0.8.0/22
20	6.0.0.160/27	35	7.0.0.128/26	200	8.0.12.0/24
14	6.0.0.192/28	30	7.0.0.192/27	150	8.0.13.0/24
8	6.0.0.208/28	10	7.0.0.224/28	100	8.0.14.0/25
4	6.0.0.224/29	4	7.0.0.240/29	60	8.0.14.128/2
2	6.0.0.232/30	2	7.0.0.248/30	6	
2	6.0.0.236/30	2	7.0.0.252	10	8.0.14.192/2
				8	
				5	8.0.14.208

	6.0.0.0/24		7.0.0.0/24		8.0.0.0/20
100	6.0.0.0/25	50	7.0.0.0/26	2000	8.0.0.0/21
30	6.0.0.128/27	40	7.0.0.64/26	1000	8.0.8.0/22
20	6.0.0.160/27	35	7.0.0.128/26	200	8.0.12.0/24
14	6.0.0.192/28	30	7.0.0.192/27	150	8.0.13.0/24
8	6.0.0.208/28	10	7.0.0.224/28	100	8.0.14.0/25
4	6.0.0.224/29	4	7.0.0.240/29	60	8.0.14.128/2
2	6.0.0.232/30	2	7.0.0.248/30	6	
2	6.0.0.236/30	2	7.0.0.252	10	8.0.14.192/2
				8	
				5	8.0.14.208/2
				9	



11.9 Structured Design

